University of South Australia

IMAGE PROCESSING AND MACHINE LEARNING FOR DETECTION OF POWER LINE CORROSION DEFECTS

MINOR THESIS

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Bachelor of Information Technology (Honours)

By

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1 Abstract

Power line corrosion defect management is an expensive and perpetual undertaking for electric utilities, since affected infrastructure can cause power outages and fallen power lines can threaten public safety. Early detection and intervention are the only mitigation strategy. Conventional manual inspections are an expensive operation, with trained specialists required on site to identify and diagnose defects. Although machine learning and computer vision techniques have been used to detect defects in power infrastructure, current techniques have not addressed corrosion detection in the power lines. This thesis proposes to use machine learning image segmentation techniques to detect and locate various types of corrosion defect in images of power lines. This thesis will create a dataset of images of corroded power lines, use the dataset to train a U-net image segmentation model, and test its accuracy on defect detection and defect location tasks. The contribution of this thesis is a demonstrated and tangible algorithm that can be adopted by any electric utility for detecting and classifying conductor corrosion defects from inspection images.

2 Introduction

2.1 Problem Domain

Vast extents of Australia's power lines infrastructure were installed between the 1950s and 1970s, and subsequently those conductors are now over 50 years into their service life. The lifespan for conductors is subject to many factors such as age, conductor type, atmospheric corrosion zones, manufacturing process, temperature exposure, load exposure, wind, vibration, and other stresses. Some conductors will reach end of life earlier than others and this makes it difficult for Distribution Network Service Providers (DNSPs) to apply effective replacement rules.

"A significant portion of the conductors installed more than 50 years ago are still in use, so conductor ageing is becoming a major concern for Australian DNSPs." (Naranpanaawe et al, 2018, p.2)

DNSP's administer comprehensive asset monitoring and replacement strategies to minimise the risk of conductor failure from corrosion (Naranpanaawe, et al., 2018). Generally, conductor inspection schedules are derived from technical knowledge about asset properties, location, age, and risk (Energex Limited, 2020). As the network ages, the risk of corrosion also increases making detection less predictable. DNSP's are also under increasing pressure to operate more efficiently hindering replacement expenditure. At current rates of expenditure on asset replacement, that is 0.5% per annum (SA Power Networks, 2019), the average overhead conductor is required to have a service life exceeding 200 years. This is not sustainable and there must be a step change in how these assets are monitored.

Figure 2 below illustrates the level of investment into overhead power lines reaching a peak during the 1950s. Although this graph does not include a vertical axis, the source caption does detail that this data is related to investment expenditure.

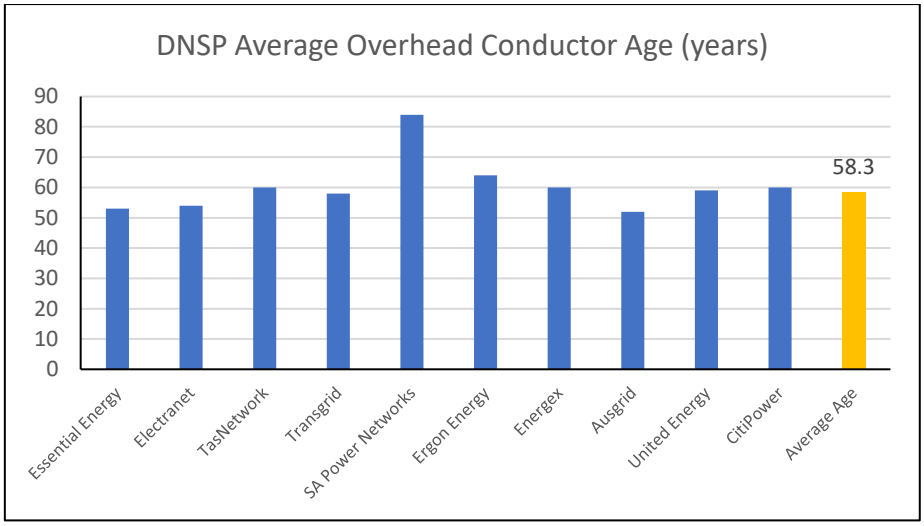


Figure 1 - Average age of conductor overhead conductors in Australian DNSP (Naranpanaawe et al, 2018)

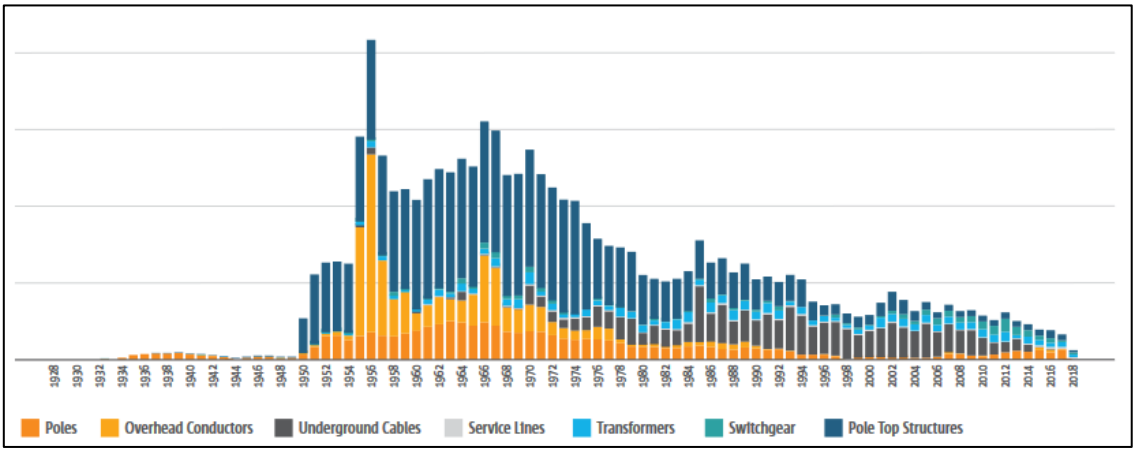


Figure 2 - SA Power Networks (DNSP) asset investment profile (SA Power Networks, 2019)

Regulatory Information (RIN) data was collated by Naranpanaawe et al (2018) and identified that corrosion is the most frequent failure mode for conductors across all Australian DNSPs as shown in figure 3. From the list of conductor failure modes, corrosion related failures are both visibly evident before failure and able to be actioned early. Therefore, visual inspections are used to identify corrosion defects.

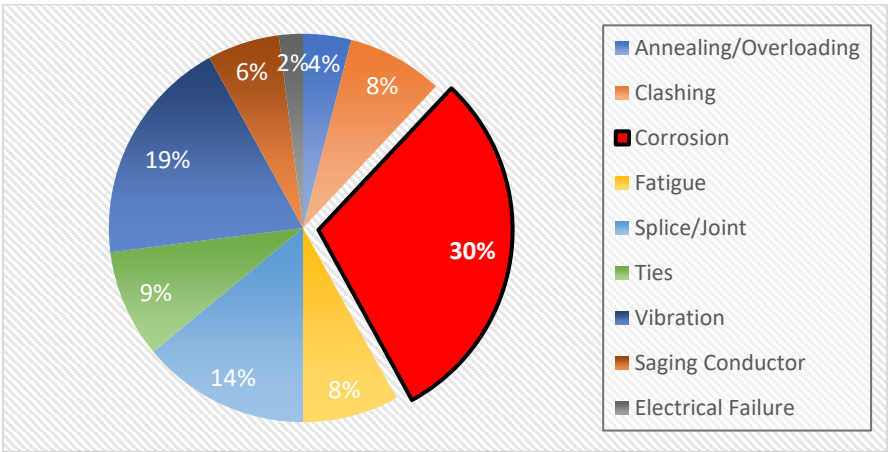


Figure 3 - Australian DNSPs conductor failure modes (Conductor condition monitoring report)

Atmospheric corrosion occurs in overhead conductors when the metals are exposed to oxygen, carbon dioxide, sulphur, chlorine, or water vapour (Naranpanaawe, et al., 2018). Galvanic corrosion is also common and occurs when metals with different electrode potentials are in contact with moisture and electric potential. The by-product of galvanic corrosion is a white powdery substance on conductor surface. Examples of corrosion defects:

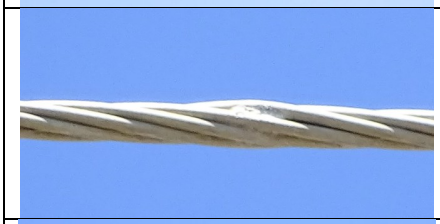
Examples of overhead conductor corrosion defects	
	Galvanic corrosion white powdery by-product
	Pitting on the conductor surface
	Conductor surface degraded
	*Bulging of conductor*
	Broken strand

Table 1 - Types of corrosion defects

For overhead conductor, the combination of age and fatigue can exacerbate the rate of corrosion (Liu et al 2020b). In 2018 a study into conductor condition monitoring (Naranpanaawe, et al., 2018) collated data from RIN documents for all Australian DNSP's and calculated the average conductor age in Australia to be 58.3 years, alarmingly, the study also found that overhead conductors have an expected lifespan of 50-60 years. This suggests that throughout the next decade the number of overhead conductor failures related to corrosion is likely to dramatically increase.

Currently, DNSPs overhead conductor management strategies rely heavily on visual inspections (Naranpanaawe, et al., 2018), and these can be prone to human factors like physical access constraints to inspect power lines, inconsistencies in visual interpretation and missed defects. Conventional visual inspection techniques are unsustainable as the vast areas of overhead conductors approach the end of their designed service life. The defects are likely to arise more frequently and expose DNSPs to greater risk. A solution is urgently needed where conductor inspection images are collected at much more frequent intervals and processed by machines that can detect accurately and consistently where corrosion has occurred.

2.2 Industry Response

A popularly endorsed solution to this problem of requiring more frequent and more detailed inspections, is to use Unmanned Aerial Vehicles (UAVs) to perform the overhead conductor inspection. The UAV industry has seen tremendous innovation in recent years. UAV sector was spearheaded by the military with early adoption paving the way for refinement of the technology for many other industries to adopt. It is now widely used in other areas of asset inspection.

Inevitably, to reduce cost, improve asset inspection quantities and safeguard the operational state of the networks, Australian DNSPs will need to move towards using UAV for frequent and effective power line inspection to keep up with the defect increases that result from aging power lines.

2.3 Proposed Solution

Industry solutions for UAV-based inspection require an operator to be present at the site to analyse the UAV footage in real time and applying specialised knowledge to identify defects. While this is effective in some special cases such as investigating the severity of known defects. This process could easily be out paced by the proposed solution described in this section. This solution entails having the capture undertaken in the field by a UAV pilot to collect imagery of many power lines. These images should be uploaded to a central system via the cloud for analysis and classification by the algorithm proposed in this thesis. The role of the skilled analyst is to analyse only the classified images.

UAV-based power line inspections will result in vast quantities of image data and specialised techniques are required for processing to be efficient. Machine learning has the capability of processing the incoming images by utilising a trained classification model from labelled images and applying this model to new data.

The specialised machine learning techniques required for analysing incoming UAV image data are investigated in this thesis, with a new algorithm that can prepare image data for machine learning is demonstrated. Machine learning algorithms are then applied and compared for suitability to this application of analysing data from the inspections.

2.4 Benefits

The benefit of providing rapid and precise power line image defect detection algorithm coupled with a transition to UAV-based inspections will result in a capacity gain for the DNSP to undertake a greater number of inspections at less cost. A more robust inspection process means less reliance on predictive replacement strategies as the actual state of the overhead conductor network can be assessed on a frequent basis. The benefits extend to the community where early detection of corrosion defects will avoid unexpected power outages and even prevent bushfires from starting.

Current inspection cycles for overhead conductor assets are 5-year and 10-year intervals depending on factors such as the bushfire risk rating of the area they exist within (Energex Limited, 2020). There is an efficiency gain associated with UAV compared to conventional inspection techniques, this

extends to a more thorough inspection capable of detecting all minor conductor defects. Considering the UAV inspection is more effective, the cycle can increase to 7-year and 12-year intervals and will translate to a 30% reduction for inspection expenditure.

The early identification of defects provides opportunity for replacement and directly translates to a reduction of unplanned outages. This can be seen as a 20%-30% saving in penalties that would otherwise be applied to the DNSP in the event of an unplanned outage, incentivising this result with a higher revenue allowance in the following regulatory period.

Utilities of Energy Networks of Australia (ENA) can expect that an image processing and deep learning corrosion defect detection model developed in this thesis will result in the following business benefits:

- Increased effectiveness of overhead conductor inspections
- 30% Reduced cycle time between overhead conductor inspections
- Reduced false negative corrosion defect detections
- Ability to process many spans worth of overhead conductor images within minutes

2.5 Research Gap

Publications that describe image processing and machine learning techniques in the application of power line asset inspection is a growing domain in recent years. A bibliometric analysis undertaken by Liu et al (2020a) shows how publications referencing 'deep learning' in the field of vision-based Power Lines Inspection has risen sharply in recent years.

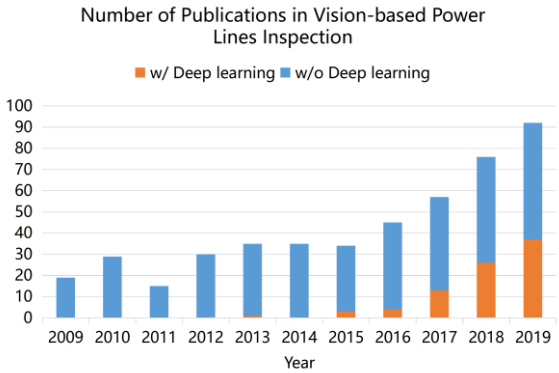


Figure 4 - Bibliometric analysis showing deep learning publications

Despite the increase in publications there are still research gaps such as how researchers can obtain real-world inspection data and with visually inspected defects included. This data is often not publicly available and held privately by DNSPs. The lack of ground-truth data in the form of power line defects make this a difficult area to research. This project will utilise a dataset of real images of conductor corrosion related defects to develop an effective image processing algorithm and machine learning model and compare results.

Many publications reference to power line inspections but use the term generally to describe all forms of transmission infrastructure. There has been significant research about transmission towers and isolators inspection with a far smaller number of published documents related to inspection of power line overhead conductor.

A conductor condition monitoring workshop was held in 2020 and Saha et al (2020) presented future project possibilities. Suggestions include using a database of labelled images of conductors along with

pre-processing of images and feature extraction techniques to develop condition assessment algorithms.

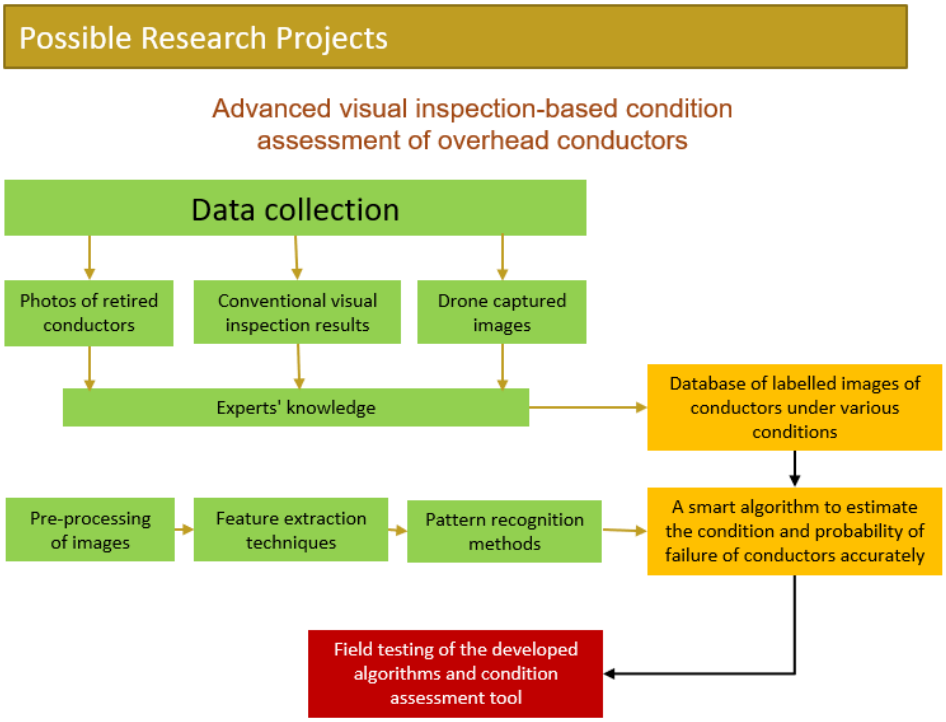


Figure 5 - Conductor Condition Monitoring Workshop - Possible research projects

2.6 Motivation

This section discusses the three factors that motivate the research undertaken in this Thesis study.

First, a transition for Australian DNSPs to utilise UAV inspections is anticipated to occur on an increasingly aggressive scale over the next few years. This will result in significant data quantities that will need to be analysed and classified, this study will result in a model that can meet this purpose.

Second, machine learning for the purpose of infrastructure defect detection is a well-documented area however for a model to be trained effectively requires large amounts of training data that includes ground truth data in the form of overhead conductor corrosion defects, which is not easily accessible to most researchers. This study anticipates using a large dataset of overhead conductor asset inspection images with defects included.

Third, where routine inspections are used by DNSPs, the analysis is performed manually (Naranpanaawe, et al., 2018). An automated processing and analysis deep learning solution that can save Australian DNSPs time, money and mitigate risk through early defect detection will be highly desirable and motivate their own interest.

3 Objective

The aim of this thesis is to investigate how machine learning and image processing can be applied to solve the problem of automated detection of conductor corrosion from inspection images. To achieve this, a dataset will be required consisting of asset inspection images with corrosion defects on power lines. To fulfil the investigation, a range of techniques selected from research will be used to build models and analyse the dataset and compare how well they perform by analysing performance parameters of the deep learning model on validation and testing data.

The objectives of this thesis are as follows:

Objective 1: Collect a dataset of asset inspection images.

Objective 2: Prepare and label the data for training.

Objective 3: Train a machine learning model to detect corrosion from processed images.

Objective 4: Evaluate how well the trained model works.

4 Literature Review

The objective of the literature review is to explore a selection of common image processing techniques that cover the main goals for image processing and to show how these are utilised in industrial applications relevant to this thesis.

This section explores articles covering a selection of image processing techniques and convolutional neural networks that are identified from literature surveys. The analysis will show how the techniques are used and how they have been implemented in industry. This is a well-documented area and an extensive range of techniques exist, usually developed to undertake a particular task with specific requirements. A collection of techniques is required for this thesis to undertake the methodology and experiments, it is necessary to identify those that will be suitable for application in this thesis.

4.1 Image Processing

4.1.1 Background

There is a substantial number of methods related to image processing for defect detection in literature. In general, they are classified into four approaches, 1) image enhancement, 2) segmentation, 3) feature extraction and 4) image classification.

In 2018, a literature survey was undertaken to categorise the assortment of published image processing techniques used in corrosion detection (Ahua & Shukla, 2017). Typical image processing process follows these main steps, with some types of techniques used in corrosion detection listed under each step:

- Image enhancement pre-processing techniques
    - Morphologic (Gaussian Blur)
    - Histogram equalisation
- Segmentation
    - Edge detection (Canny edge-detector)
    - Threshold-based
    - Region-based (Colour space-based detection)
    - Clustering
- Feature extraction
    - Wavelet Transform
    - Gaussian / Fourier
    - Gabor filtering
    - Texture analysis
    - Gray Level Weight Matrix
    - Histogram of Gradients
- Image classification
    - Gans
    - Neural Nets

4.1.2 Techniques

Gaussian Blur

In image processing, noise is the random disturbances in image signal caused by external sources (Swain, 2018). Image acquisition is always subject to the presence of noise, therefore any techniques that can mitigate its impact are desired. The Gaussian blur is one such technique, a convolution of the image using a kernel of a certain shape has the effect of smoothing image values. The kernel enumerates each image pixel and establishes a new value as the weighted average of that pixel's neighbourhood.

Histogram equalisation

In grayscale images, intensity frequency of pixel values can be plotted to a histogram. An equalisation acts to spread the intensity range across the values. The effect of this can reveal areas of low contrast by providing a higher contrast (Sudhakar, 2017).

Canny edge-detector

Edge detection is achieved by calculating magnitude and direction of image gradients via a kernel to extract those pixels forming edges within an image. The Canny edge-detector specifically extracts edges at a one-pixel width, making it unique from other implmentations such as the Sobel operator where edges of varying width are returned (Rong, et al., 2014). For this reason, Canny edge-detector is well suited to engineering applications where precision is required.

Threshold-based segmentation

This technique is used to differentiate pixels into classes by calculating histogram intensity values to apply as the threshold (Dhanachandra & Chanu, 2017). The *global thresholding* technique is a common technique although it is sensitive to background noise and subtypes of this algorithm may be used if the image is subject to imperfect illumination. *Adaptive thresholding* is proposed by Yand and Chen (1994), where Gray-level variation across the image is calculated first and adjusts the histogram values during processing. This technique can also overcome background noise and has been used successfully in power line segmentation from aerial-based images with background noise (Zhang, et al., 2019).

Because the images are ground based pictures of overhead power lines, the sky colour consistently forms the background and will need to be removed before performing the feature extraction methods. The *global threshold-based segmentation* algorithm will be used. The aim of this technique is to isolate those pixels forming the power line (foreground) from the background class of the image.

The *Otsu method* (Ng, 2006) is a *Global thresholding* technique suited to bimodal images where there is a foreground object and a background. The *Otsu global thresholding* technique is simple and therefore efficient thresholding algorithm (Fenyi, et al., 2020).

Clustering

Clustering is an unsupervised algorithm used in image segmentation. Groups are formed based on similarity in the values divided into a specified number of groups (Chauhan, 2019). The aim of clustering is to extract objects within the image. K-means clustering is a popular technique because it is simple and effective.

Gabor filtering

Texture analysis and edge detection can be achieved using the feature extraction technique Gabor filter. It is a convolution filter that combines both a Gaussian component and a sine component. The Gaussian provides the weights and the sine provides the directionality (Bhattiprolu, 2020). The Gabor filter permits a certain 'band' of frequencies and rejects the others.

Grayscale conversion

Selecting a suitable RGB to Grayscale transformation algorithm does have an impact on performance relative to the objective of subsequent processing. Kanan and Cottrel (2012) identify thirteen different methods for colour-to-grayscale and compared results showing that the selection of method does indeed have an influence on performance. Kanan and Cottrel (2012) suggests Luminance for texture recognition is an important characteristic to preserve in grayscale conversion. It is preserved by the most common techniques utilising a weighted mean of the RGB image channels.

Wavelet transforms

A multi-resolution wavelet analysis tool is shown to be an effective tool to segment corrosion areas from the rest of an image (Mohammad, 2009). For pre-processing, this technique offers efficiency gains due to elimination of redundant background noise. To segment corrosion-like regions from the rest of the image, both texture and colour analysis are applied. Multi-resolution wavelet analysis is a powerful tool to characterise features for texture classification.

Colour space-based detection

A methodology is proposed in (Medeiros, et al., 2010) for corrosion detection using colour and texture descriptors to classify corroded surfaces on carbon steel storage tanks and pipelines. The study reveals that colour changes on metallic surfaces are often related to corrosion. Colour is shown to be an important feature for corrosion detection systems. In the proposed models, hyperspectral imaging statistics (HSI) were used to characterise the colour of metallic surface samples on images from optical cameras. HSI is a technique to divide images into many bands beyond the visible wave lengths humans can see (red, green and blue). Hue, saturation and intensity can be derived from these channels and objects or regions of interest (ROI) such as corrosion produce a unique 'fingerprint'.

Gans (Generative Adverserial Network)

A Gans network consists of a generator model and a discriminator model, both are usually a type of neural network. The generator learns how to replicate the data it sees in the training set, its objective is to 'fool' the discriminator network, by continually improving until the data is as good as the training data. The discriminator determines whether the data is real and is data from the validation set, or a fake from the generator (Goodfellow, 2017). An example use is to generate a larger dataset from a small dataset.

N-Nets (Neural Network)

A N-Net is a deep learning algorithm comprising of node layers with an input and an output layer. Each node in the structure can be activated to pass data through to the next layer when a given weight exceeds a threshold. Data that arrives at the output layer contributes the result of the classification (IBM Cloud Education, 2020).

"*The goal of morphology operations is to simplify image data, preserve essential shape characteristics and eliminate noise*" (Goyal, 2011, p. 161).

Image segmentation

The objective of image segmentation is to segment an image such that classification is output at a pixel-based level. Segmentation can be achieved with numerous techniques, Pal and Pal (1993) find that the selection of segmentation technique is directly related to the types of images and the application area. The authors find that literature for colour image segmentation is not as well documented as grayscale segmentation. Neural network-based image segmentation is very promising with the potential to segment images in real time.

4.2 Convolutional Neural Networks

4.2.1 Background

A substantial number of images is usually required to train a convolutional neural network (deep learning model). For example, in a previous study to locate conductor in UAV images (Liu, et al., 2020), a dataset of 6000 images was used to build a deep learning cGAN (Douzas & Bacao, 2018) model. Similarly, in (Liu, et al., 2020) a texture-based image processing model to detect conductor was built using over 4000 images of conductor.

Convolutional neural networks (CNNs) are popular for their superior performance with a wide selection of input data. Fundamental elements of a CNN are the convolution layer, pooling layer and fully connected layer (IBM Cloud Education, 2020).

In the convolutional layer, a feature detector (kernel) parses the image and performs a transformation by application of a filter. The kernel checks to see if a feature is present. This process is a convolution. In a CNN a feature is an edge, when the edge is detected that detection acts as a weighting on the node, if the weight exceeds the threshold of the node, then it is activated and passes the data through to the next connected layer. In the final layer, the activated node represents a classification that the model has predicted.

Image segmentation by neural networks uses convolutions to process small areas of an image to detect minor features in the form of edges. The outputs of the model are combined to label an area of the image. The advantage of neural networks is their ability to define rules based on labelled data and apply these rules to new data.

The U-Net architecture is also reviewed in this section, it is a modern neural network that is capable of efficient image segmentation, and it is an immensely popular model in the biomedical industry. U-Net is a suitable tool for this thesis as it is exceptional for performing image segmentation. This processing will require efficiency, according to the U-Net official website – a 512 x 512 size image will take under a second to process. The other key reason for choosing to use U-Net is because it can be trained on very minimal number of images.

4.2.2 Techniques

Computer-Vision technique

A weakly-supervised Convolutional Neural Network (CNN) is demonstrated by Lee (2017), to visualise localisation of powerlines to mitigate some of the safety issues for UAVs. This algorithm can assist with UAV navigation. One challenge that presents differently from conventional ground-based inspection of power lines is the irregular background noise. In this algorithm, noise was filtered out by applying image processing such as first order derivative of Gaussian.

Augmentation

Data augmentation refers to techniques used to increase data quantities by replicating existing data but with slight modifications so that for the purposes of machine learning it appears differently. Mikotajczyk and Grochowski (2018) Address the problem of class imbalance within datasets by using data augmentation. Images of smaller class sizes can have images duplicated with adjustments to the copied images applied. This can achieve a class balance. Some examples of techniques include affine transformations such as zoom in, rotate and shear. Another popular technique is colour based, such as contrast adjustment, histogram equalisation and white balance.

4.2.3 U-Net

The U-Net model consists of an encoder and decoder path that pass the image data through convolutional and max pooling layers (Ronnenberger, et al., 2015). An encoder and decoder exists on every layer in the U-Net model. The encoder is responsible to reduce the spatial dimensions and increase the channels, this process is called max pooling. The decoder performs the opposite function. Each layer also has a skip connection, it is responsible to preserve some information from the encoder path. This preserved data is later used to add spatial context during the decoder process on the corresponding layer. The skip connections are denoted as the grey horizontal arrows and the encoder decoder paths are respective red and green arrows in figure 6.

The number of layers determines how many feature maps (channels) are generated at the deepest part of the model, known as the 'bottle-neck'. For example, if there are four layers with 32 filters and input images of dimension 128x128px, at the deepest layer in the U-Net model there will be 512 feature maps generated with dimensions for each of 4x4px. Decoding the feature maps to reconstruct the original dimensions involves concatenating the data from skip connections with input data from the previous layer and up-sampling.

What makes the U-Net model unique from traditional convolutional neural network models is the technique of using skip connections to map between the input and output pipelines. The skip connections concatenate output of the feature map from the corresponding layers encoder with the input from the previous layer's decoder block. See the horizontal arrows in figure 6. The skip connections are advantageous in this model because they provide a spatial reference for the incoming data to the decoder block, giving more precise segmentation results.

"The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization." (Ronnenberger, et al., 2015, p. 234)

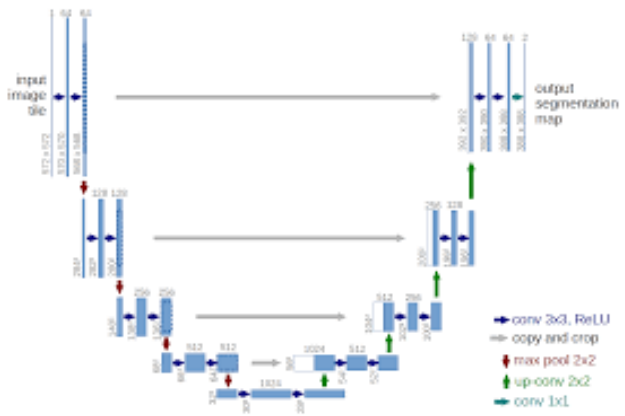


Figure 6 - U-Net architecture with skip connections (Ronnenberger, et al., 2015)

4.2.4 Training Metrics

This section describes some common metrics for image segmentation.

Intersection Over Union (IoU)

IoU is an evaluation metric commonly used for semantic segmentation (Jordan, 2018). The calculation is the number of pixels overlapping between the ground truth and the prediction, divided by the area of union of the predicted and ground truth pixels. It is commonly used for semantic segmentation because it rewards true positive predictions for achieving a good overlap with the ground truth. This is a suitable measure for minority classes such as broken strand, however it is poor metric for dominating classes such as background.

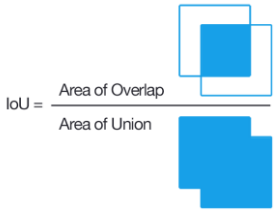


Figure 7 - Intersection over Union

Dice Co-efficient

This metric is popular as a loss function for image segmentation, it measures overlap between samples (Jordan, 2018). The Dice coefficient described in the equation below is based on a binary mask for A and B (i.e. where values above a threshold, e.g. 0.5, are interpreted as 1 and the values below are zero). The soft dice coefficient uses the probabilities emitted by the model, multiplied by the binary mask, instead. This soft variant is more easily differentiable than the thresholed variant, hence it is often preferred as a loss function. The soft dice loss is then 1-(soft dice coefficient).

$$Dice = \frac{2\,|A \cap B|}{|A| + |B|}$$

Figure 8 - Dice co-efficient [12]

Precision and Recall

In image segmentation, **recall** is the measure of '*proportion of boundary pixels in the ground truth that were successfully detected by the automatic segmentation'* (U Toronto, n.d.).

For **precision**, the calculation is the '*proportion of boundary pixels in the automatic segmentation that correspond to boundary pixels in the ground truth'* (U Toronto, n.d.).

4.3 Industrial Applications

4.3.1 Image Processing

Image processing objectives generally fall into four categories:

1. Checking for presence
2. Object detection and localisation
3. Measurement
4. Identification and verification

Object detection and localisation as a form of inspection is commonplace in many industries, examples include woven fabric inspection, roads, pipeline, storage tank, steel bridge inspections. The techniques are embedded into automated systems with well trained and very effective models. Implementations of image processing for defect detection and a selection of techniques specific to corrosion detection are reviewed.

Bridge stay-wire corrosion detection (PCA Analysis / Feature Classification)

An image processing algorithm is successfully implemented by (Ho, et al., 2013)**.** It can detect cable faults on bridge cable stay-wires from images captured by a cable traversing robot. Key features in the images are highlighted using image enhancement techniques median filter and histogram equalisation to effectively adjust contrast. Then, PCA analysis is applied to identify the main directions of variation in the data. PCA Analysis suppresses dimensions that contribute little to the overall shape of the data. When compared to the training data, returned value provides the damaged or undamaged stay-wire image classification.

Pavement crack detection (Otsu Thresholding)

An unsupervised method is developed by Akagic et al (2018), for efficient image segmentation of pavement crack detection utilising Otsu Thresholding technique. The algorithm presented in this paper uses the Otsu Threshold value and deducts the most frequent gray level value to establish the least frequent gray scale value to eliminate values with a high ratio. The result of this algorithm is less noise around cracks in the images because more frequent values surrounding the cracks are marked as background pixels. The algorithm performed well with an accuracy score of 97.13%, however the F-Score was 76.09%. This is still a good result when compared to other implementations detailed in the paper. The method was tested on a dataset consisting of only 50 images selected from a Google Images search, citing a lack of publicly available datasets.

Classification of Basmati rice grain variety (PCA Analysis)

An algorithm to detect Basmati rice varieties utilising the PCA Analysis is proposed by Kambo and Yerpude (2014). Images are captured of Basmati rice grains using a camera and placing the rice on a black paper. Median smoothing is applied to the image to remove noise and unlike other smoothing techniques (Gaussian or averaging), median smoothing protects the edges from eroding. Edges are very important to determine the Basmati variety as the grain length is a precise feature. PCA is used to establish features of individual rice grain samples per image, five features are extracted for each sample including: area; length; eccentricity and perimeter. This data is used to build a statistical model for classifying each Basmati variety. Overall accuracy of this model was 79%, this figure is very accurate because of the high availability of sample data.

Tile defect detection (grayscale conversion analysis)

Ceramics production is a multistage process, with six categories of defect possibly developing. The most frequent defect is the crack which is caused by rapid temperature changes. Detecting cracks was achieved by using structural algorithms such as Morphology and Edge detection. Blob is a type of patch defect; an example is drops of water on the tile surface (Karimi & Asemani, 2014). Blob detection was achieved through use of statistical models including histogram curve, co-occurrence matrix. Pinhole defects could be found using wavelet transform and Gabor filter, but this was computationally expensive meaning it was not real-time.

Jean fabric defect detection (Optimized Gabor Filter)

An improved algorithm utilising optimised Gabor filtering is proposed by Ma et al (2020) for the purpose of defect detection of denim fabric. Utilising a Gabor filter bank consisting of four scales and six directions, the algorithm selects the optimal filter by using two-dimensional entropy evaluation. Average detection rate was 91.25% compared to 86.75% for the non-optimised version of the algorithm.

Road cracks defect detection (grayscale analysis and Gabor)

An important aspect of road safety is the management of road defects, certain types left untreated can even be fatal. The major issue with image processing for this purpose was "*the presence of noise and randomness in the shapes of the defects*" (Chatterjee, et al., 2018). A solution was required to identify road defects accurately and efficiently while driving over them. One implemented solution was a feature extraction algorithm using super pixels to group similar areas (pixel intensities) (Mouselimis, 2021). Grey Level Co-occurrence Matrix (GLCM), Gabor and Histogram of Oriented Gradients (HoG), Local Binary Patterns and other algorithms were used in various crack type detection with high accuracy achieved (Chatterjee, et al., 2018). The study found texture-based features were highly effective at crack and defect detection.

Fabric defect detection in textiles (pattern recognition)

Manual inspection of woven silk fabrics is a very fatiguing process for a trained inspector and automation is a proven solution (Sabeenian, et al., 2012). The complexity in this inspection process is to build algorithms to detect the different types of possible defects (15 in total). The study found that Gray Level Weight Matrix (GLWM) was effective for detecting holes and stains. Multi Resolution Combined Statistical and Spatial Frequency (MRCSSF) was able to detect minor defects such as missing yarn and broken filament. Accuracy of the classification model using sliding window object detection, was 100%, however the dataset consisted of only 130 images of size 512 x 512.

Vision-based broken strand detection for power-line maintenance robot (Histogram of Gradients)

In power line fault detection, malfunctions of the overhead ground wire (OGW), is one of the most serious. The OGW is the lightning protection for extra-high voltage power lines, also used for high-speed protection communication between substations, without this wire large areas of the grid are vulnerable to outages. Also, a broken OGW strand can dangle in proximity to the extra high voltage conductors and cause flashover.

The method involves extracting the image gradient feature from the region of interest (the OGW) (Song, et al., 2014). In the case of a normal OGW, there are two principal orientations a vertical orientation of the edges of the OGW, and second, the winding helix angles across the wire. In the case of a broken strand, there is a different configuration that will cross the usual vertical and helix

orientated gradients. The broken strand will always have a lower gradient intensity for the vertical and helix gradients. This forms the basis for classification.

In the image pre-processing stage, the grayscale image is equalised to show the strands clearly and reduce the effect of lighting conditions.

Using Static Vector Machines (SVM) to compare a known normal OGW HoG to new images, the model can classify normal, broken strands and obstacles on the OGW. An accuracy above 96% was achieved by using this algorithm. In figure 9 and 10 below, the HoG outputs are shown with a greater variance across the histogram for the image with the broken strand.

Although this algorithm is shown to work well, the requirements for this implementation involve installing a robot on the OGW to traverse individual spans. This may not be feasible for wide-scale adoption because of the variation of obstacles for a robot, eg. Insulators and line fault indicators. The robot weight on the line could also cause damage. Providing evidence that this practice can achieve the same results from ground-acquired imagery could be valuable for electric utilities.

A normal OGW and the HoG output:

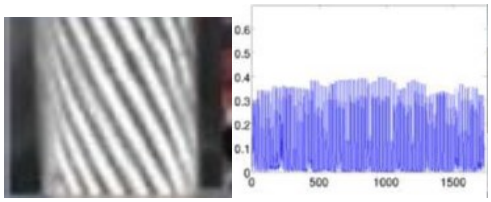


Figure 9 - HoG for normal OGW

A broken strand OGW and HoG output:

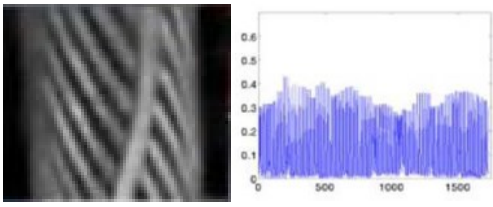


Figure 10 - HoG for broken OGW

4.3.2 Convolutional Neural Networks

Railway line surface defects detection (Convolutional neural networks)

A defect detection method is proposed by Shang et al (2018) for localising and classifying defects on rail images. CNN is used to extract rail surface features from cropped rail images (from video footage) and divide them into two categories, 'in-tact' and 'defective'. Recall levels are improved on previous existing algorithms, while also reducing the calculation complexity and training time. CNN model 'Inception-v3' model (a branch of GoogleNet) is used to reduce computational complexity and allows the rail images to be spatially converged without losing feature information. The experimentation models were trained on 5793 images with 466 defective rail images, finally tested on 1517 images with 91.1% accuracy and 95.36% accuracy. The algorithm lacks real-time performance, however and

requires further performance optimisation work. In this work, a lack of available defect images was cited, and a up-sampling method was used to synthesize more minority class samples.

Pipeline surface condition assessment (Convolutional neural networks)

A neural network for analysis of vibro-signals to classify the defect type is proposed by Vankov et al (2020). Two main types of defects that are detected in pipeline assessment include crack-like defects and corrosion defects. Conventional monitoring involves Fourier transform analysis on acoustic signals, however this is not suitable for non-linear and non-stationary signals. The model 'ResNet V2' is implemented using the Python programming language with Tensorflow framework. The model was trained utilising a dataset of 44100 signal values, derived from 340 defect-free measurements, 1020 'hole' measurements, 277 'crack' measurements. The authors achieved an accuracy of 99% achieved on the test dataset.

Power transmission tower detection (Convolutional neural networks)

In this paper, Want et al (2019) develop a deep learning model to detect power transmission towers from UAV images. Detection and localisation of the towers is considered a prerequisite for the future of UAV automatic power transmission tower inspection. Two convolutional neural networks models were compared for this task, 'Faster R-CNN' and the 'YOLO-V3'. The results show that 'YOLO-V3' is the optimum performer for real-time detection, while the 'Faster R-CNN' model obtained a better detection accuracy.

Defect detection method for substation equipment (Convolutional neural networks)

On the topic of substation equipment inspection, this paper proposes a solution to the "*prominent surge in (substation inspection) workload and the relative shortage of personnel*" (Ying, et al., 2020, p. 6318). A convolutional neural network model for detecting defects in substation equipment is developed using the 'Faster R-CNN' model. A selection of 16 common engineering-defects are detected in a dataset of 15,000 substation defect images. The authors prescribed a strategy to overcome a problem of a heavily skewed dataset, whereby many samples belong to only a small number of categories. The solution applied during image pre-processing involved data enhancement operations to make varied copies of less sampled classes, so that an even distribution of image sample categories is achieved.

Insulator detection using image augmentation

Deep learning with convolutional neural networks has proven to be effective for such tasks as insulator recognition (Lee, 2017). Augmented images were derived from an original 3000 images dataset. A six-layer CNN model was trained to perform the detection. Sliding window method was used along with a line-fitting method to precisely place bounding box.

4.3.3 U-Net

U-net for mining land cover classification based on high-resolution UAV imagery

In the fight against deforestation, the frequent mapping of land coverage every five years is not sufficient for the mining areas in Central Highlands region of Vietnam (Giang, et al., 2020). The authors propose a UAV for frequent data acquisition in the form of multi-spectral imaging. A U-net model is trained to detect six land cover types. The study assesses the advantage of deep learning with high-resolution images for land cover detection in mining areas. The trained U-net model achieved an accuracy score of 83%, making it a valuable tool for the mining managers. Output from the U-net

model is shown in figure 11 - in this output image, clearly make out water bodies in blue, bare soils in grey, mining land in yellow and grass land in bright green.

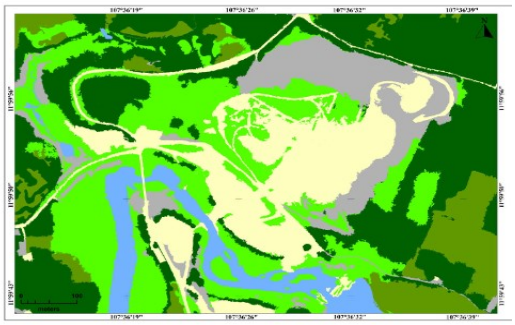


Figure 11 - U-Net for mining land cover

Road Extraction by Deep Residual U-net

A combination of the strengths of residual network and u net are combined in a hybrid 'ResU-net' (Zhang, et al., 2018). It is compared against other deep-learning methods. The test is to detect roads from aerial imagery. The 'ResU-net' model was shown to perform better at road extraction than all other deep learning methods including better than U-Net.

Convolutional networks for biomedical image segmentation (U-Net architecture)

A new architecture for convolutional neural networks is demonstrated in (Ronneberger, et al., 2015), called U-Net. The architecture addresses requirements of the biomedical image processing for localisation, whereas conventionally these models only address image classification. In the output, each pixel should be given a classification (semantic segmentation).

The U-Net architecture is proven to outperform the previous best method for segmentation (sliding window convolutional network). It is shown to require fewer training images and can operate much faster (a 512x512 image takes less than one second). In figure 12, results of the ISBI cell tracking challenge, where U-Net won the segmentation challenge with 92% Intersection on Union (IoU).

"U-Net has been a remarkable and the most popular deep network architecture in the medical imaging community, defining the state of the art in medical image segmentation (Ibtehaz & Rahman, 2019, p. 4)*"*

U-net architecture was published in 2014, a google scholar search for "U-net" now yields over 60,000 publications.

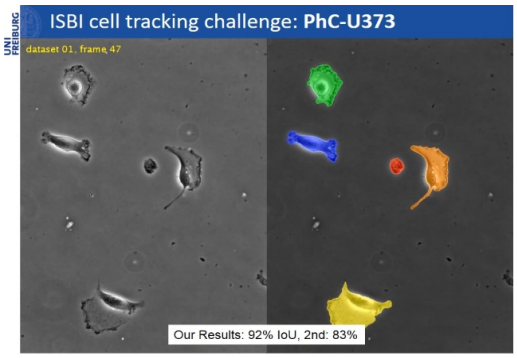


Figure 12 - Award winning U-Net performance

5 Gap Analysis

There is some research presented in the literature review that shows how defects are identified on power line infrastructure (Song, et al., 2014); (Ying, et al., 2020); (Lee, 2017), such examples include transmission towers and substation equipment, however there was no research found relating to the specific methods for image processing for detection of corrosion on power lines, that this thesis will address.

6 Research Questions

Image processing techniques have been used broadly, this thesis will focus on similar applications and propose an algorithm for use with detection of corrosion defects in images of power lines.

RQ1: which image pre-processing techniques should be used to allow training of semantic segmentation on minimal data?

RQ2: How well does the U-Net model perform at defect detection in ground-based overhead conductor images?

RQ3: How is the U-Net performance affected by other factors like image resolution and the number of annotated images?

7 Methodology

Developing an image processing algorithm for autonomous corrosion defect detection in power line images, will achieve the objective of this thesis. Developing the algorithm involves training a U-net model, that is trained and evaluated using a collected image dataset. The model is trained using custom loss and selection of metrics, and the results of the model are compared with some other training experiments. There are three major phases in the method: 1) data collection and 2) image pre-processing and 3) model training and evaluation. Each phase is described in corresponding subsections. A high-level overview of the methodology is presented in figure 13:

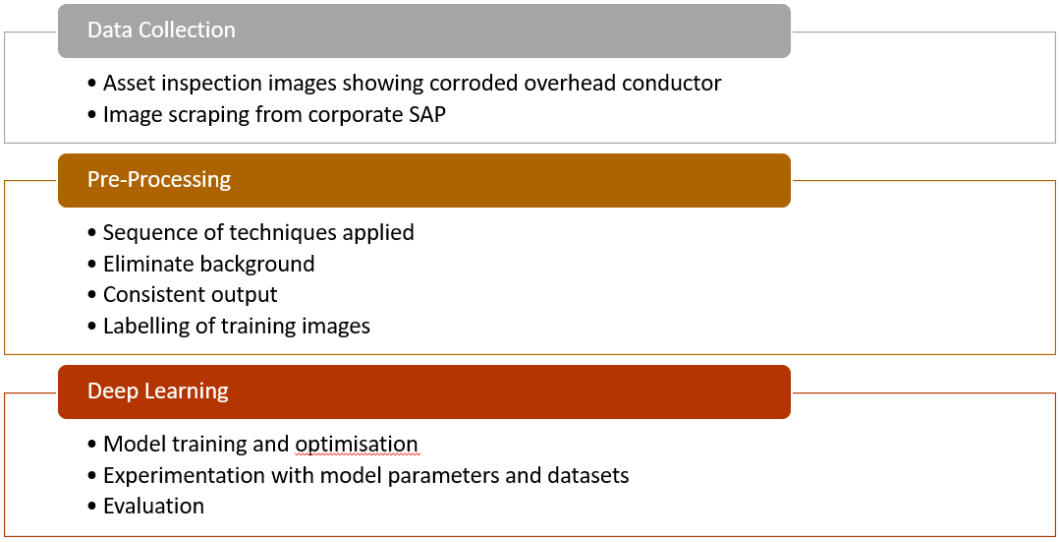


Figure 13 - Methodology overview

Phase 1 - Data Collection

At first, an image dataset is required; the images are sourced from a DNSP corporate SAP system with permission. Specifically, the data comes from attachment images in SAP notifications where corrosion affected power lines have been identified during routine power line inspections. Scripts are developed and used to copy the images from the SAP repository into this image dataset. Training data is selected and used to explore image processing techniques and for training deep learning models. Image labelling is undertaken in phase 2, as the collected images are not labelled.

Phase 2 - Pre-Processing

In this phase, a sequence of image enhancement, feature extraction and segmentation techniques are applied to each of the training dataset images. The power line in the image is oriented across the image, and the background is eliminated so that deep learning in phase 3 is effective.

Experimentation is recorded and those techniques that generate a consistently high-quality output become the answer for **research question 1**.

All images from the pre-processed training dataset are labelled to identify and outline the corrosion defects. In this process it is important to be precise and consistent in the technique so that the model training is effective.

Phase 3 - Deep Learning

In this phase, a U-net model is trained using the labelled images. The model is trained using a custom loss function and with appropriate metrics for an imbalanced dataset. The result of initial training is benchmarked for further experiments on learning rates, image scaling and varying quantities of dataset images. The performance assessment from these experiments is used to answer **research question 3**.

8 Data Acquisition and Processing

8.1 Data Collection

Image data stored in corrosion defect notifications are collected via image viewing web-apps available in the corporate environment. Python web scraping tools are applied to collect the images from JSON response to GET requests, the scraping tool runs for half an hour to collect this dataset of images.

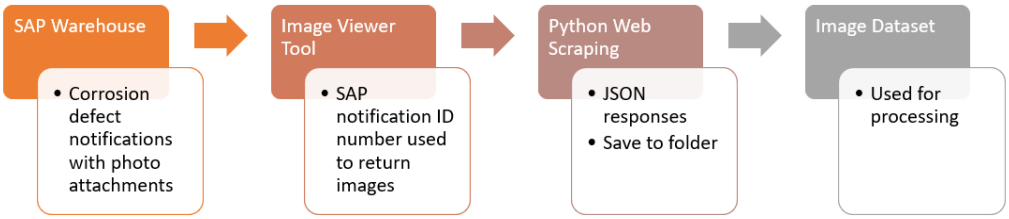


Figure 14 - Data collection overview

The SAP warehouse is a repository of notifications raised by asset inspectors for replacement or monitoring of power line corrosion defects identified during power line inspections, the common notification field indicates the 'corrosion' related defect. Photos from the inspection are uploaded as attachments to the SAP notification. Access to this data source is valuable for assessing performance on images that have been captured during real overhead power line inspections.

Images that are uploaded into SAP are available via a corporate asset image-viewer web-app via the notification number. The image scraper code is used to download the images from the asset image viewer application, making use of the API (Application Programming Interface) and JSON response.

The images vary substantially when considering the dimensions, conditions, and general quality. They have been collected over some years using different cameras and photographers, more recent photos are of higher quality. The quality of all photos collected from SAP warehouse are subject to the JPEG compression applied during the uploading process.

The nature of this collected data is that each image will contain a corrosion related defect. Defects do not take up many pixels in the images, subsequently a class imbalance problem is presented (all images contain a high proportion of background pixels), this is a problem for deep learning models and will need to be addressed in the pre-processing and deep learning phases.

There are approximately 6000 images in the dataset, representing over 4GB of data. The images show power line segments containing defects. Within the collected data, there is only limited metadata to derive a classification table to establish the composition of defect types within the whole dataset. In the images filenames there is a clue to what defect is present, however this is not a mandatory field, see figure 15 showing that majority of images have no evidence of the classification of the defect determined from the filename.

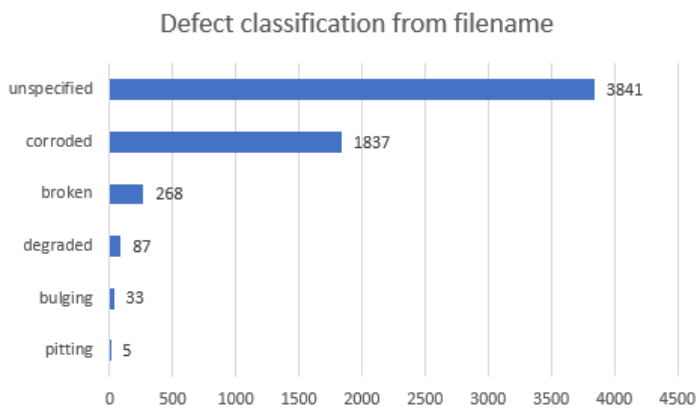


Figure 15 - Metadata classification

The filename data suggests an initial list of corrosion related defects to undertake classification for.

In figure 15, there is evidence of a class imbalance problem with a high proportion showing only corroded images (galvanic corrosion). The images from this dataset to be used in deep learning training will need to be selected with care so that all defect classes and their various conditions are represented.

The following sample images from the dataset show the types of corrosion defects:

Corrosion defect type	Description	Sample image
Galvanic Corrosion Build-up	White patchy areas between the conductor strands show a build-up of galvanic corrosion deposit	
Broken strand	A broken strand fused to the corroded conductor	
Pitting	Dark patchy areas appearing on the outer most surface of the conductor. Pitting is distinct evidence of subsurface corrosion (D&D Coatings, n.d.)	

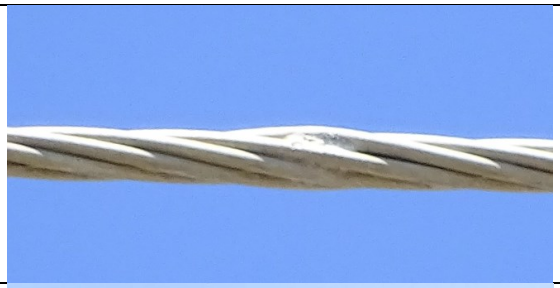
Bulging	Galvanic corrosion deposits around the central steel core have expanded the cores	
Degraded	Corrosion has caused high impedance to the point where the structure has melted	

Table 2 - Classification of defect types

8.2 Pre-Processing

The images are pre-processed so that they are consistent in appearance, and to encourage effective results from the deep learning training. The image pre-processing is undertaken to address four key problems with the properties of the images as they exist in the data collection state.

1. Power line is present in any orientation across the image
2. Highly imbalanced proportion of background pixels vs foreground power line pixels
3. Image dimensions are too large to be compatible with deep learning
4. Various lighting conditions and other background objects and with date stamps

The images are high resolution with an abundance of background pixels. This is not ideal for deep learning semantic segmentation where all pixels in the image will need to be processed and classified and there is nothing to gain from processing background pixels, affecting performance. To address these problems, a pre-processing algorithm is proposed to perform the following sequence of techniques on each image.

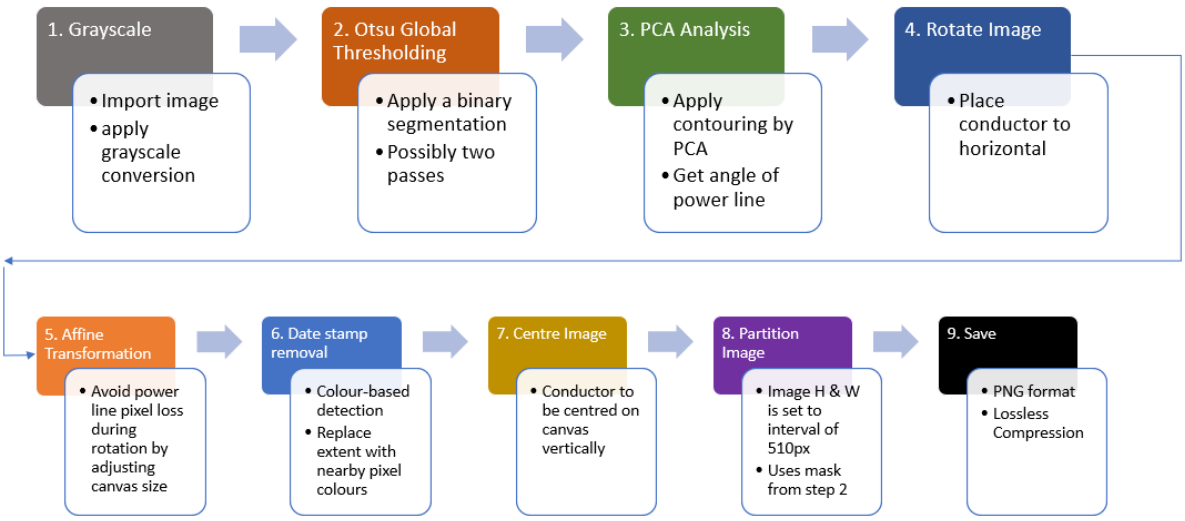


Figure 16 - Pre-processing overview

Conversion to grayscale

To simplify the analysis on this image in preparation for thresholding, the three channels are reduced to a single channel. The conversion technique is to convert the BGR channels into HSV and then separate out only the Hue channel.

The Hue channel is used to handle the common sky-blue background colours in the images. It has the effect of converting them into a narrower band of values than other grayscale conversion techniques.

Otsu Thresholding

Thresholding is carried out in two passes. In the first pass, pixels with value lower than the threshold value represent foreground and inversely greater values comprise the background of the image. Foreground is given a value of 255 and background a value of 0.

After the first pass, the pixel count for white pixels and black pixels are compared, if white pixels is greater, it is likely that abnormal sky conditions are present. Occasionally, a portion of the sky is detected within the foreground, usually this occurs when cloudy conditions are present. Only in this situation does the thresholding get re-run in a second pass.

In the second pass, the brightness attribute from the HSV colour space, “V”, is used to convert the original image to grayscale. Re-running the Otsu thresholding using this brightness colour space has the effect of separating the brighter clouds from the darker conductor clearly.

PCA Analysis

When the Otsu Thresholding value is applied to the image, the output is a binary segmentation. This is input to the PCA analysis module to contour the feature areas (those pixel areas with values of 255) and return an array of eigenvectors. The direction of the largest eigenvector represents the orientation of the powerline in the image, which is subsequently used for roation and centering.

Image orientation

Handheld inspection photos result in images with conductor of varying angles, rotating the image so that the conductor is consistently horizontal is demonstrated in studies (Liu, et al., 2020). The technique implemented to calculate rotation is the eigenvalue feature data from the PCA Analysis.

Otsu Global Threshold-based segmentation results

The thresholding technique works well for most cases, occasionally the power line has a range of luminance values both above and below the threshold, see the example in figure 17:

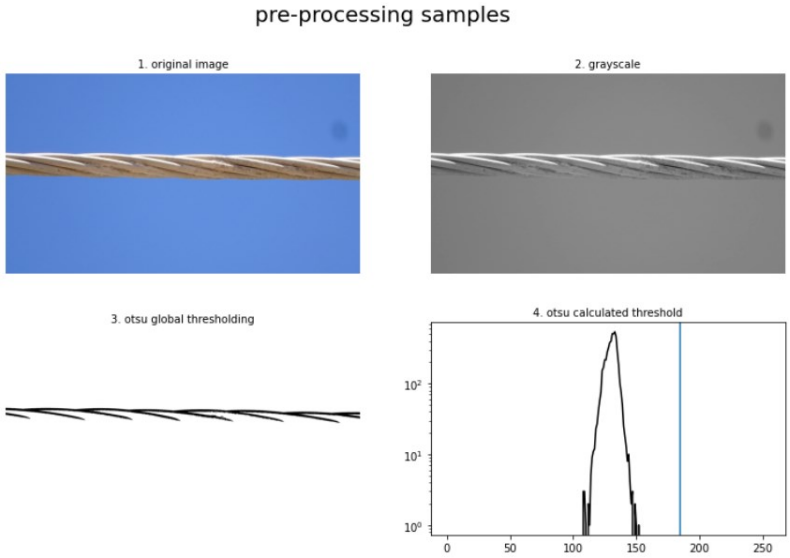


Figure 17 - Otsu thresholding sample

The thresholding time to run results are included in table 6.

Approximately 8 of these images have a processing time of about 1 second, the remainder are less than 0.5 seconds. This can be explained as occurrences where the output has a higher count of white pixels than black pixels, so the second pass of the Otsu Thresholding is activated, in the second pass, the brightness pixel values are used rather than the hue pixel values used in the first pass. This has the effect of detecting power line pixels that are shown darker than the background. See in the table 3, the original image is shown with a cloudy white background:

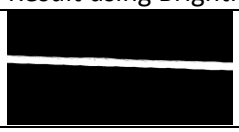
Original Image	Result using Hue	Result using Brightness
		

Table 3 - Otsu thresholding iteration result

Image date stamp removal

Many images in the dataset have a date stamp embedded. This presents a problem where the value of those pixels would always fall within the threshold value during 'Otsu Thresholding' and they would be segmented into the foreground along with the powerline. This is a problem because the date stamp was visible at completion of all pre-processing. The solution is achieved by selecting nearby pixel values to replace those pixels with a RGB value of the precise colour of the date stamp. Because of the JPEG compression, often the surrounding pixels around the date stamp text were affected, so by getting the top left and bottom right most pixels matching the RGB colour, a box area of pixel colours was able to be copied from nearby pixel values.

The technique adjusts the image to remove the date stamp as shown in table 4:

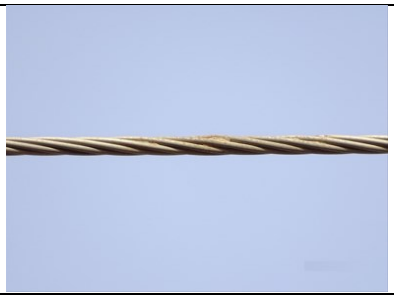
Before	After
	

Table 4 - Date time stamp removal sample

Output before date stamp handling	Output after date stamp handling

Table 5 - Date time stamp removal results

Regarding the performance impact of this step, testing the removal of date stamp for 10 images took on average less than 1 second.

Pre-processing overall performance

A performance test was undertaken to assess the time taken to process 20 images from the dataset selected at random.

The time test results:

	Processing Time (s)				
Image ID	**Otsu Global Thresholding**	**Date Stamp Time Removal**	**Overall Processing**	**Output dimensions**	**Output file size**
1	0.45	0.74	11.19	5120 x 2048	30MB
2	0.32	0.59	9.56	5632 x 1024	17MB
3	1.49	1.58	5.08	5632 x 512	8MB
4	0.41	0.56	10.55	5632 x 512	8MB
5	1.02	1.53	8.03	5632 x 512	8MB
6	0.35	0.51	7.96	5632 x 512	8MB
7	1.16	1.18	7.22	5632 x 512	8MB
8	0.56	0.73	10.59	5632 x 512	8MB
9	1.05	0.61	7.6	5632 x 512	8MB
10	0.53	0.57	12.5	5632 x 2048	34MB
11	1.05	0.85	5.11	5632 x 512	8MB
12	0.47	0.67	14.45	5632 x 1024	17MB
13	1.11	1.81	6.71	5632 x 512	8MB
14	1.06	0.64	9.62	5632 x 512	8MB
15	0.47	1.75	10.03	5632 x 512	8MB
16	0.48	0.58	10.44	5632 x 512	8MB
17	0.57	1.35	8.99	5632 x 2048	30MB
18	0.95	0.83	9.07	5632 x 512	8MB
19	0.54	0.7	7.03	5632 x 512	8MB
20	0.57	0.65	8.01	5632 x 512	8MB
TOTAL	**14.61**	**18.43**	**181s**		**248MB**

Table 6 - Pre-processing run time

The results indicate an average pre-processing runtime of 9s per image. The mean output size is 8MB.

The output dimensions in table 6, result from the rotate and crop function where the image size is at first reduced to the extent of power line pixels, and then increased (buffered out with pixels of value 0) as the image is rotated. The rule for this script is to increase dimensions to the next interval of 512 to prevent any pixel loss of the power lines. Some of the images have vertical dimensions up to 2048, in this case the power line pixels in the original image after rotation take up between 1536 and 2048 vertical pixels in the original image.

8.3 Labelling

The labelling process is a manual task undertaken in this thesis; 400 images are labelled to identify four defect classes. Labelling consistency and precision is pivotal to the success of the deep learning phase. When the labelling is complete, a python script is used to convert all images into a new data structure whereby the image, mask and json file are output. The mask image shows each class represented by a colour and background class is black. The images are then carefully separated into training, testing and validation sets.

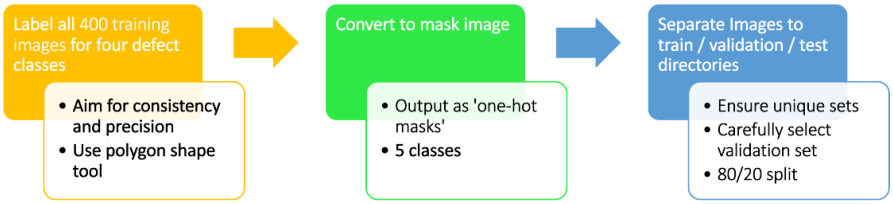


Figure 18 - Labelling overview

Labelled data is prepared to mimic the output desired from the model. It provides a ground truth for the model to train against and it enables metrics to precisely monitor how well the model is performing by assessing at a pixel-based level the quantity of correct and incorrect predictions.

The masked images represent the classes according to the colour. It is important that this colour to defect type index is consistent throughout all training datasets. An example of a labelled image with classes associated by colour is provided in table 7.

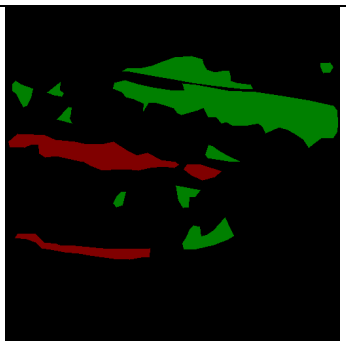
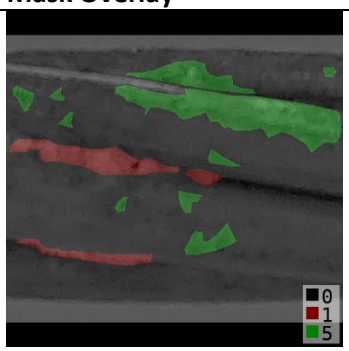
Original	Mask	Mask Overlay
		

Table 7 - Masked image example

All 400 of the images used for model training are labelled to show four different classes listed in table 8:

Examples of overhead conductor corrosion defects	
	1 - Galvanic
	2 - Degraded
	3 - Broken Strand
	4 - Bulging
	5 - Pitting

Table 8 - Classification of defects for labelling

In the interests of simplifying the training and to reduce the complexity involved with classifying 5 classes, bulging (no.4) defect type is excluded from the labelling process.

In table 8, galvanic corrosion defects are the easiest to label, as they stand out brightly contrasting the remainder of the image. Broken strand defects are similarly well-defined, with vertical lines crossing perpendicular to the power line and the helix pattern is clearly disrupted.

Pitting is the most challenging defect for labelling as there is often many dark patches of random shapes across the face of the power line. Degraded conductor takes on different shapes and sizes, it may be present in some images as an area of substantial pitting, classifying the difference between pitting and degraded conductor is challenging.

The image labelling is undertaken on the 'label-me' software package included with Anaconda desktop installation. In the software, a directory is nominated, and it is then possible to iteratively process each image and export a corresponding json file that shares the same name as the image as well as the coordinates and class name of each polygon vertex.

With only 400 images being labelled from the original set of 6000 images, it is important that such a small training set are labelled with a high level of precision and consistency. Establishing this level of consistency is with practice and becoming familiar with the appearance of defects in the images.

9 Deep Learning

In this section details of the development of a U-net model are presented, including how training is undertaken, experimentation and evaluation.

The pre-processed images train a model that can be used to predict corrosion defects on any power line image. This method aims to guide development of the model with incremental improvements. An initial model will be developed with minimal code, the results of this model establish a baseline against which subsequent models are assessed. In the next iteration, multi-class segmentation is implemented with output showing the plotting of predictions.

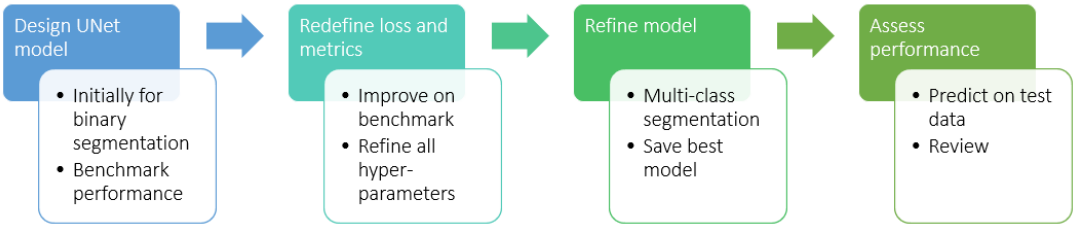


Figure 19 - Deep learning overview

9.1 Initial Model Development

As a preliminary experiment, a simpler binary segmentation model is developed to inform the design of the main segmentation model. A basic level of training is achieved for a single class of defect, galvanic corrosion, with a binary segmentation metric guiding the optimisation.

A four-layer U-net model is implemented with 32 filters per layer and a binary cross entropy loss function used for optimisation. Training is performed for 50 epochs. This initial model shows that the data format is compatible with training and predictions are generated. Inspection of the predictions shows a poor precision of the defect class, where the predictions often exceed the boundaries of the ground truth.

The initial model proves that the labelled images are suitable for training and that the predictions exhibit poor precision. The cause of poor precision is likely due to not enough training, and a class imbalance where the images are comprised mainly of background. In the main segmentation model, this class imbalance requires attention.

9.2 Training

The metrics and loss from the preliminary experiment are redefined to accommodate the imbalanced properties of the dataset. Hyper-parameters such as epoch number, batch and step sizes are adjusted to improve performance. The learning rate and additional class weights are applied. Input characteristics are accommodated, namely image size and labelled classes. Finally, the best performing model is assessed using predictions on test data.

The training is undertaken using the U-Net architecture similar to the preliminary experiment, to achieve multi-class image segmentation model using a highly imbalanced dataset of conductor corrosion defect images. The validation data was hand-picked to ensure that each condition occurs within it. Details of the model and training process are described here.

To begin with any processing a training dataset is selected, 400 images are removed from the original dataset and divided into three groups for the purpose of training the model. The first group of images is the training dataset, this represents 225 images. The second group is for validation and is made up of 22 images. Then 43 are used as test data. The remainder of the 400 form the dev data.

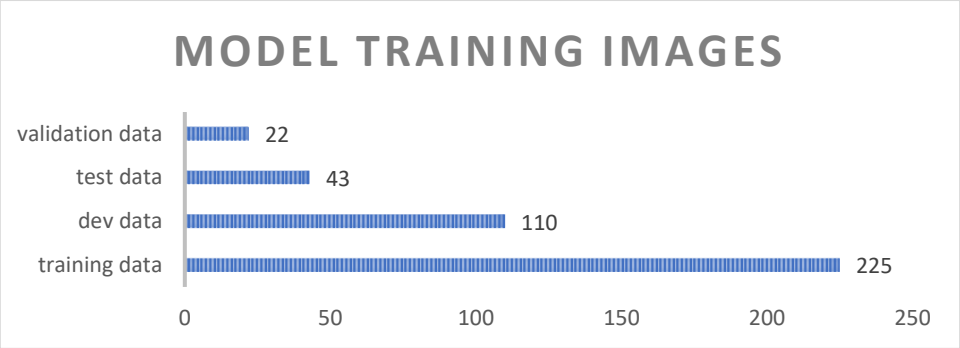


Figure 20 - Train test split of data

Dataset preparation

The 400 images to be used for model training are divided into three unique sets and stored in separate directories. The training data and the validation sets are used to fit the model, while the test set is the unseen data and provides the realistic performance measure for how the model predicts on new data. It is important in this train test split operation to not allow any images to exist in any more than one of the sets as this can mislead the training performance.

The input shape for the model is 128x128x3, this is different to the output of the pre-processing images that are 512x512x3. The reason detailed in the results section, the images were down sampled prior to the train-test split to eliminate the need for partitioning images into patches as would usually be required for processing images of size of 512x512x3, as the memory requirements at this size are too great for the GPU processor.

Training metrics

Before undertaking the model training, consideration is given to the dataset and how the significant class imbalance may affect key metrics including the loss function. In the dataset there is a low frequency of defect data when compared to the background pixels in the images, to the extent where >90% of dataset image pixels are background. Comparing the non-background classes in the dataset shows a similar bias of >70% towards galvanic corrosion defects. This bias must be factored into the loss function, so that similarly to (Hashemi, et al., 2018) and their imbalanced medical image dataset, false negatives must be treated with more importance than false positives. Recall measure is favoured in the model training because it does not penalise the returned false positives, recall does look at the number of false negatives.

Typically, for an imbalanced dataset, the IoU or Dice coefficient function would be used, however in this dataset the imbalance is much too great for either score to be effective. To equip the loss function to not simply ignore the minority classes, a custom Dice coefficient is used, with different weights applied to the losses obtained for each class. Classes with a higher weight are more likely to be predicted. In the case of the majority class, background, an f_beta weighting of 0.5 is applied so predictions in this class trade precision in favour of recall.

Class weights are assigned to prevent the dominant class of background from skewing the classification process. To calculate the class weights to be applied in the loss function, each class is weighted inversely proportional to the square of the pixel frequencies per class. The background is many factors more frequent than the other classes, so the weight applied to background is 0. The class weighting results are demonstrated below:

Br = Broken Strand; Ga = Galvanic Corrosion; De = Degraded Conductor; Bg = Background

Class	Br	Ga	De	Bg
Calculated Weight	0.958	0.014	0.028	0.00

Table 9 - Class weighting technique results

The weighting is applied at the sigmoid operation during the learning process to balance the dominance of the background class. In the results above, the first class is highly unlikely to be classified as the dataset only contains 4681 pixels of this class. This applied weighting increases the likelihood of candidate pixels being classified as this minority class.

U-net model

The model is constructed using customisable U-Net architecture provided by Tensorflow Keras U-Net model.

U-Net model parameters

The U-Net model used for training has been assigned the following parameters:

Parameter	***Value***
Input shape	128x128x3
Activation	Rectified Linear Unit (Relu)
Number of classes	4
Filters	4
Number of layers	4
Use batch normalisation	True
Upsample mode	Transposed convolutions
Dropout	0.3

Table 10 - U-Net model parameters

Model Training

The model training is calculated over 400 epochs and every epoch comprises 23 steps and a batch size of 10. This means that 10 training images are processed at every step to utilise the 225 training images, after each batch of 10 images are processed, the metrics are updated to monitor the performance. The trained model is then inferred on all images from the validation dataset and key metrics such as the IOU (Intersection over Union) and the validation loss then guide how the training process is adjusted to minimise the loss and maximise the IoU on the validation data. Training is optimal at the epoch achieving the lowest validation loss score. The model at this epoch is saved out to disk so that it can be used for inference on other data. The training time for this network was approximately 30 minutes on an NVIDIA Tesla P100 GPU.

Using the custom recall favoured loss function as the measure of training performance, the loss curve established convergence in figure 23:

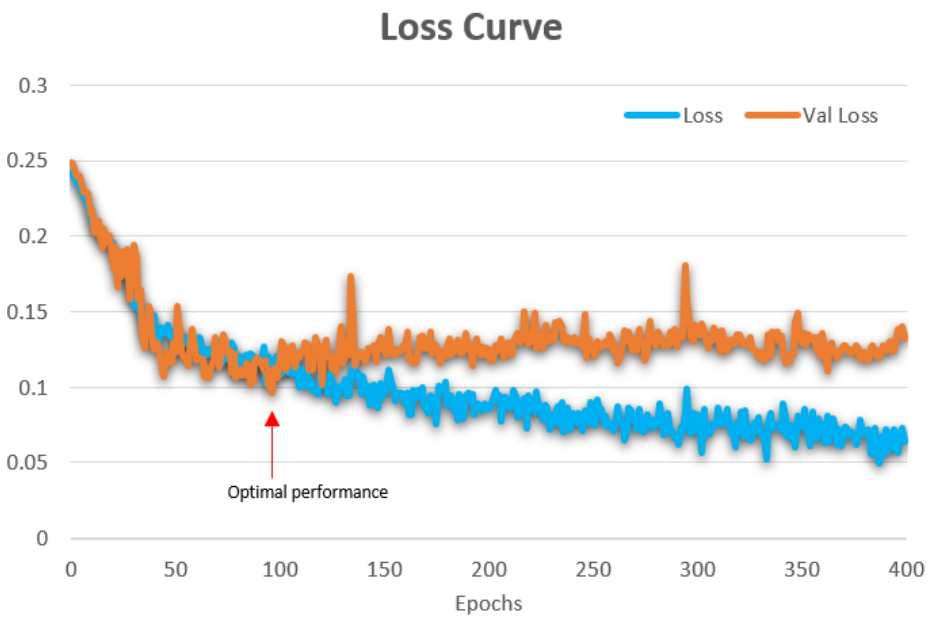


Figure 21 - Deep learning loss curve

Optimal performance on the validation data was reached at epoch 97 with a validation loss of 0.096. At this point, a model snapshot is taken and used to assess the performance on test data.

Regarding other metrics from the training, the IoU score reached a peak of 0.3 on the validation data in epoch 40

U-Net training parameters

Parameter	***Value***
Epochs	400
Batch size	10
Call-back	Save model to file
Call-back monitor	Validation loss
Call-back save best only	True
Call-back save weights only	True
Optimiser	Adam
Learning rate	Custom cyclical learning rate
Loss	Custom loss (recall-favoured and class weighted)
Metrics	IoU

Table 11 - U-Net training parameters

The model training progress is governed by a custom defined cyclical learning rate. The learning rate is an important hyper-parameter for tuning deep neural networks. In (Smith, 2017) a new method called cyclical learning are applied to this parameter. Cyclical learning is shown to be effective for the trial-and-error phase of typical machine learning process. In the usual case, a machine learning model learning rate is a specified value that takes several attempts to set correctly. Until that ideal value is found, the convergence phase of the training will either take too long or fall into a local minimum. In

this case, the optimisation has converged to the local minimum, and is unable to reach the global minimum. The cyclical learning is designed to vary cyclically within a band of values. The cyclical learning rate was applied in this training and the parameters are described below:

Parameter	Value
initial_learning_rate	0.000001
maximal_learning_rate	0.01
step_size	5
scale_fn	1 / (2.0 ** (x - 1))
scale_mode	Cycle

Table 12 - Cyclical learning rate parameters

In the initial epoch, the learning rate is set low, and the maximum permitted improvement is set to 0.01 to ensure the learning rate remains steady. The improvement in this case is measured by the recall favoured loss parameter taken at every fifth step in the epoch, as per the authors in (Smith, 2017) this value should be updated between 2x and 8x the number of training iterations per epoch (batch number). The scaling function governs how steadily the learning rate is increased.

9.3 Training Experiments

In this section, parameters of the U-net model are adjusted to determine their effects. The objective of this process is to obtain the best results.

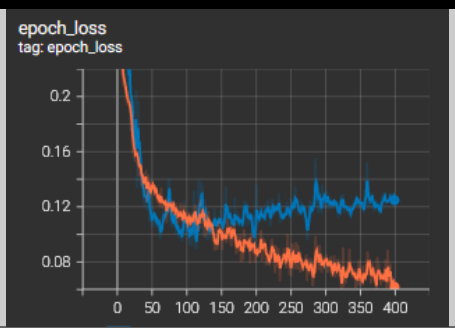
Experiment 1 – Learning Rate

Learning rate is a scaling factor that determines how much change in the weight to permit. The scaling prevents the weights in the model from changing too much. Learning rate influences the speed of learning and may have an effect on how closely the model can get to the optimum. A large learning rate may mean faster learning but may be unable to reach the optimum. In the U-net implementation, a cyclical learning rate is applied where the parameters determine how often to cycle between maximal and base (initial) learning rate. The benefit of using a cycle is that optimisation can escape local minima situations and subsequently find the actual convergence.

The aim of experiment 1 is to evaluate how changes to the learning rate parameter can reduce overfitting and underfitting to maximise generalisation. Loss curves that exhibit an optimal fit have three main features (J, 2013):

1. Training loss plot stabilises at a point
2. Validation loss plot stabilises at a point
3. The gap between the two curves is minimal

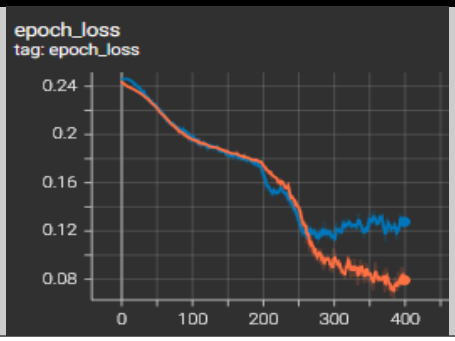
The following results are the notable changes identified while adjusting the learning rate. The model and details are presented, followed by a short description of the outcome. The parameter changes from the previous model are highlighted in yellow.

Model ID	Learning Rate Parameters	Loss Curve
01		

parameters	values
initial	1e-3
maximal	1e-2
scale	1 / (2.0 ** (x - 1))
step-size	5
optimal loss	0.096
Curve trait	overfitting

Table 13 - Experiment 1 model 1

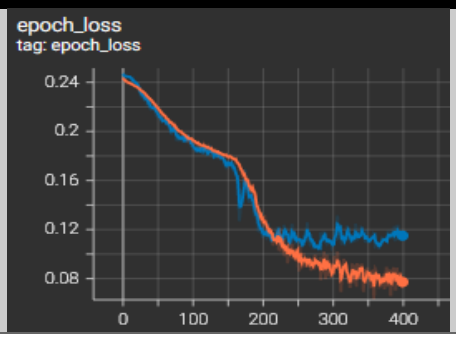
Model ID 01: The maximal learning rate results in a model that shows some overfitting, in the loss curve the orange training loss continues to improve, while the validation loss peaks just before epoch 100 and deteriorates out to epoch 400.

Model ID	Learning Rate Parameters	Loss Curve
02		

parameters	values
initial	1e-4
maximal	1e-3
scale	1 / (2.0 ** (x - 1))
step-size	5
optimal loss	0.10369
curve trait	overfitting

Table 14 - Experiment 1 model 2

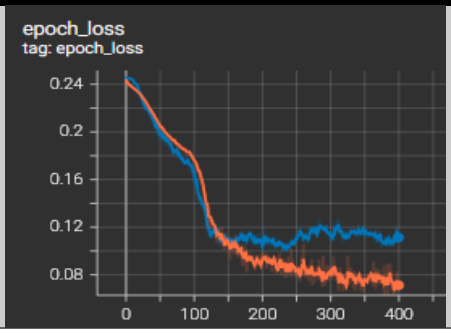
Model ID 02: In the previous model, the learning rate was shown to be too aggressive, in Model 02, the initial and maximal parameters are reduced. In the Model 02 curve the optimal loss is only discovered at epoch 300, so the learning rate is slower, although there is a flatter validation plot in epochs 300 – 400, the training plot continues to improve.

Model ID	Learning Rate Parameters	Loss Curve
03		

parameters	values
initial	1e-4
maximal	1e-3
scale	1 / (2.0 ** (x – 2))
step-size	5
optimal loss	0.10739

Table 15 - Experiment 1 model 3

Model ID 03: The scale parameter is reduced slightly; this appears to allow greater peaks in the learning cycles. The validation curve shows a plateau beyond epoch 200, although the training plot continues to improve. The slower learning rate has not improved optimal loss from Model ID 02.

Model ID	Learning Rate Parameters	Loss Curve
04		

parameters	values
initial	1e-4
maximal	8e-3
scale	1 / (2.0 ** (x – 1))
step-size	5
optimal loss	0.09962

Table 16 - Experiment 1 model 4

Model ID 04: The scale parameter is turned back, and the maximal parameter is increased given the promising results in Model 03. This has the effect of lengthening the plateau on the validation data, additionally an even lower optimal loss discovered at around epoch 250.

Experiment 1 conclusion:

Some noise in the plotting is visible in all loss curves, this is the evidence of the cyclical learning rate oscillating between max and min, in the initial curves where the learning rate permitted is much greater there is also more noise. Overall there was consistency in the optimal loss between all models at a score near 0.1. Minor improvements were obtained. The optimal loss score was best for model ID 01, however, the curve of model ID 04 exhibits a more stable validation plot, where the validation plateaus and does not exceed 0.12. The selected training rate parameters are those of model ID 04.

Experiment 2 – Training and Validation Dataset Variations

The aim of experiment 2 is to evaluate the IoU score training performance using different quantities of training and validation datasets. The learning rate parameters are determined from experiment 1. The dataset sizes used to train the model in experiment 1 are detailed below and this is the baseline performance:

- Training: 225 - Validation: 22 – Test: 45

In the results table the IoU is presented for each class, the abbreviations are:

Br = Broken Strand; Ga = Galvanic Corrosion; De = Degraded Conductor; Bg = Background

Test ID	Training Size	Validation Size	Test Size	Validation Set - IoU results			
				Br	**Ga**	**De**	**Bg**
TD1	225	22	45	**0.25**	**0.41**	**0.10**	0.96
TD2	180	22	94	0.19	0.35	0.1	**0.97**
TD3	120	22	160	0.01	0.33	0.02	0.81
TD4	80	22	204	0.01	0.08	0.02	0.97

Table 17 - Experiment 2 results

Results for experiment 2 show that better performance is achieved when the training dataset is larger. TD1 had significantly higher detection rates of the minority class broken strand, though only a minor improvement on the galvanic corrosion class. Broken strand class is the least frequently occurring in the training and validation datasets, therefore the drop is easily influenced when one is removed from the dataset. Ga is the most frequent class in both datasets and this explains why the score is only impacted when the dataset size is smallest. Background detection is largely unchanged throughout

the results, due to its dominance in each dataset. Degraded conductor class is the worst performer. This may relate to the how the two labelled classes are merged (pitting and degraded). Initially the class merge was performed to achieve a balance, both have similar characteristics in colour, though the shape difference is significant. This result may indicate that the shape difference is cancelling out any training on this class. Degraded conductor is also the most challenging class to label accurately.

These results suggest that changes to the dataset could benefit training such as increasing the number of images with broken strand defects and finding images where the degraded conductor is clearly shown.

Experiment 2 conclusion:

Overall, the initial 225 images used for training achieve a reasonable IoU score on broken strand and galvanic classes. The dataset size is large enough to get results and there is evidence of diminishing returns as the size of this dataset increases from TD4 (80) to TD1 (225). Each image is important in this set as they were selected specifically to represent examples of each class.

Experiment 3 – Image Scaling

Using the learning parameters from Model 04 in experiment 1 and the image datasets from experiment 2, a benchmark is established to undertake experiment 3. In experiment 3, the scaling factor is changed to observe how the image dimensions can influence performance.

In the benchmark model, the trained images are of the dimensions 128 x 128 x 3. This is the result of applying a scaling factor of 4 on the pre-processed images where the native image size is 512 x 512 x 3. Initially this scaling down was implemented to reduce the memory demands on the GPU as resources were depleted. Now with further scaling this experiment will record how the performance changes.

Test ID	Scaling Factor	Image Dimensions	Validation Set - IoU results			
			Br	**Ga**	**De**	**Bg**
SF4	2	256 x 256 x 3	0.27	0.34	0.08	0.97
SF1	4	128 x 128 x 3	0.25	0.41	0.10	0.96
SF2	8	64 x 64 x 3	0.18	0.4	0.08	0.97
SF3	16	32 x 32 x 3	0.07	0.18	0.13	0.96

Table 18 - Experiment 3 results

Experiment 3 conclusion:

Adjustments to scaling factor provided interesting results. Notably, increasing scaling to a factor of 8 in SF2 increased IoU score for broken strand and degraded conductor in the test set.

This could be explained as the poorly labelled test images (where labels tend to overlap defects) having pixel counts reduced per labelled defect due to scaling down.

Scaling factor of 16 in SF3, is too great as the image data is over simplified and the rate of missed defects increases.

SF4 provides more intricate segmentation because of the increased dimensions, it does mean that training time is longer.

Scaling down images does influence the training performance. It has the effect of increasing efficiency of the processing, and it is also detrimental to the performance of the model. Getting this scaling to the right size is key to balancing available resources and getting the desired results.

9.4 Results on Test Dataset

When the optimised model is inferred on the test dataset of 45 images, the results are promising for broken strand and galvanic corrosion detection when viewing the prediction masks. This does contrast with the metrics that show IoU scores of 0.23 and 0.22 respectively. The score for galvanic corrosion does not align with that achieved with the validation data during training – validation score was 0.41.

Test Set - IoU results				
	Broken Strand	**Galvanic Corrosion**	**Degraded Conductor**	**Background**
Test	0.23	0.22	0.08	0.92
Validation	0.25	0.41	0.10	0.96

Table 19 - Results on test dataset

Using IoU as the performance measure, a score above 0.5 is considered as a good prediction (Rosebrock, 2016). However, in the test data there is another factor that should be considered with this score, the labelling quality. The most precise labelled data was selected only for the training datasets and with the remainder used as the test dataset. This is mainly the initial labelling work and generally overlaps the defects by some margin, contributing to the poor IoU scores. An inspection of the test dataset predictions reveals a more intricate prediction that is within the bounds of the labelled area. For example, see appendix A image 4 showing galvanic corrosion, the predicted areas are entirely within the labelled extent (white pixels), also the model has predicted some degraded defects precisely. This result highlights the importance of using precise labelling data throughout all datasets.

Regarding the performance on degraded conductor defect type, the result is consistent with that of the training datasets with a 0.08 IoU score and a 0.1 achieved in training dataset. This is a relatively poor detection rate, and it is likely due to compromises made in the training, labelled dataset has four classes of defect, while for training only three are used – the degraded defects and pitting defects are merged to simplify the training processing. The defects have similar properties such as darker pixels surrounded by lighter pixels, although the degraded areas tend to be much larger in shape. This result shows that merging of classes in the process, is not ideal for training.

For the broken strand detection, there are many cases where the image edge is visible due to the image processing algorithm, the detector predicts these frequently as broken strand. In practice the end of the conductor does not extend to the image boundary due to roation preprocessing. This long trailing edge represents all common traits that broken conductor exhibits. One technique that was used to reduce these false predictions was increasing the penalty on precision with a higher weight in the beta assigned during the training. This had the effect of applying a higher penalty when trying to predict areas larger than what is has been trained to detect. This did not eliminate the problem, but did reduce the detection there slightly. This is because the conductor edge has the same appearance as broken conductor class. An alternative is to repeat the last non-black pixel of the conductor to the end of the image. Another option is to crop the image back so that the conductor is always flush to the end of the image.

Examples of effective **degraded conductor** detection in Appendix A 12.1: Images 1, 4.

Examples of effective **broken strand** detection in Appendix A 12.1: Images 3, 15.

Examples of effective **galvanic corrosion** detection in Appendix A 12.1: Images 1, 4, 25, 35.

10 Discussion

In the pre-processing phase of this thesis, techniques were selected that would work together to produce the desired result of eliminated background pixels and no loss of power line pixels. The algorithm works very well in the most frequent conditions in the dataset where blue sky forms the background, and the power line is illuminated clearly. Consistent output can be observed in Appendix A, all original images show consistent orientation of the power line, and it takes up maximum pixels within the image. These results are promising; therefore, this technique combination provides an answer to the research question 1: ***Which image pre-processing techniques should be used to allow training of semantic segmentation on minimal data?*** Those techniques providing high quality results are:

- Gaussian blur to reduce noise across the image
- Otsu thresholding
- PCA Analysis
- Rotate Affine
- Custom techniques for image partitioning, background removal and date stamp removal

This combination provides sufficient results, although more customisations can be done to address where the output was poor such as in the presence of poor lighting, background objects and excess noise. In the case of the dataset, these factors were present only in a minority of cases.

In phase 3 of the thesis, the deep learning training was undertaken using custom loss function and metrics used to evaluate the performance of the model. Model optimisation was performed for 400 epochs and the results favoured a loss score of 0.096. Using this score as a benchmark, further experiments were undertaken to assess how hyper parameters and other training variables can affect the model performance. This assessment measures how changes of hyper parameters and training dataset/image sizes affect training a model, and this answers research question 3: ***How is the U-Net performance affected by factors like image resolution and the number of annotated images?***

- Image resolution directly affects processing time, as scaling an image to smaller dimensions means less pixels to process and therefore a faster training time.
- Image resolution also affects the precision of predicted pixels, IoU score was found to drop when images were both scaled up and down from the baseline scale of 128 x 128 x 3.
- Number of annotated images has the largest impact on minority classes, such as broken strand and degraded conductor. The IoU score dropped significantly for each interval of less images.
- Learning rate is shown to impact heavily on the loss curve, by changing the cyclic max and min, the rate of optimisation was directly impacted, at the right setting the convergence was greatly improved so that evidence of plateau is seen on the validation plot.

The U-net model was used in this thesis because in literature it is shown to be effective with minimal training data, and it is an award-winning deep learning model for semantic segmentation. The results are evident in the prediction assessment on the test dataset. The model detects some classes such as galvanic corrosion better than the other minority classes. The metrics provide the answers for research question 2: ***How well does the U-Net model perform at defect detection in ground-based overhead conductor images?*** The U-Net model predicts the classes at the following IoU scores for validation and test data:

Br = Broken Strand; Ga = Galvanic Corrosion; De = Degraded Conductor; Bg = Background

Validation dataset - IoU results				Test dataset - IoU results			
Br	**Ga**	**De**	**Bg**	**Br**	**Ga**	**De**	**Bg**
0.25	0.41	0.10	0.96	0.23	0.25	0.08	0.92

Table 20 - Results of U-Net performance

In these results, galvanic corrosion is predicted at a good level in the validation data, with a lower score recorded against the test dataset. A likely factor for this being the quality of the labelled data in the test dataset, earliest labelled data was used in this set and overlapped the corrosion defect by some margin. Contributing to a poor IoU score, where the precision of the prediction is actually better than the ground truth.

Although metric results are a good indicator, it should be considered in addition to the actual predictions in the images, using those in the appendix A. There are predictions occurring for each class and they are highlight intricate and often more intricate than the labelled data that they are being compared to in the IoU scores.

The results demonstrate a good level of detection is achieved, with predictions generated for all defect classes. Galvanic corrosion defects achieves the highest detection rate, with broken conductor also being detected well. Degraded conductor detection was poor, this is due to the merging of two similar classes pitting and degraded. While they are similar in colour, the shape difference and variance of conditions is too great to achieve good training levels. Additionally the pitting labelling was usually very intricate and therefore compressed out by the scaling down applied to the images. Broken conductor detection requires further work, in most cases this class is detected well – except where there is conductor truncated within the image, in this case it resembles the broken conductor.

Future Research

In future research, attention should be provided to increasing the number of precise labelled images to reach a complete representation of all images in the dataset. Increasing the labelled images is expected to yield a better performance. The pre-processing algorithm can be improved with greater tolerance to background noise and other objects. If many more labelled images are available it may be possible to forego the pre-processing steps and segment directly from the original images. Additionally, the use of alternative image processing technique such as the Canny-edge detector may be capable of direct feature extraction to identify broken lines such as in the presence of a defect. With more robust pre-processing, application can be extended to use with UAV captured imagery.

The end-to-end processing is undertaken on multiple software packages, for example, pre-processing occurs in Anaconda-Spyder software, while deep learning undertaken on Jupyter Notebooks. With a model saved to disk from the training, it is possible to combine all of these operations into one software package, utilising a drag and drop image collection will make this algorithm very appealing for other users.

With additional meta data, the ability to further analyise defects to establish the probability of failure, a rating on the severity and even prediction of the remaining life of the conductor. With geographic coordinate meta data, a list of locations could be generated of exactly where corrosion defects were identified. These techniques could also be applied to video footage, to run the algorithm frame by frame.

11 Conclusion

In this thesis, a new approach to corrosion defect detection is tested. Using image pre-processing techniques and a contemporary deep learning U-net model with tuned parameters, an effective detection level is achieved for three classes of defect – broken strand, galvanic corrosion, and degraded conductor.

A dataset of 6000 images was collected and a selection of 400 images was used for labelling and training. 400 epochs of training were undertaken on a carefully calibrated U-net model, with an optimal loss score of 0.096 achieved.

In the data acquisition and pre-processing phases, suitable techniques were identified and implemented that resulted in images that are well prepared for deep learning training. The design of pre-processing techniques addresses how the power lines in the image are oriented and achieves a consistent appearance for most dataset images. The design also balances how power line pixels are preserved while maximising the background pixels removed.

The deep learning phase successfully trained a model that detects and classifies the corrosion defects on power lines. Metrics used to assess and build the model are designed to carefully support the nature of the pre-processed images. Class weights are assigned to address class imbalance, and a customised loss function designed to optimise the segmentation.

The images used for training testing and validation were re-arranged and experiments conducted with different numbers of training images. Then model parameters were changed to explore and record how the model performance was affected. Scaling image sizes down reduces processing cost, and this was shown to affect the model performance.

Regarding the research questions, ***which image pre-processing techniques should be used to allow training of semantic segmentation on minimal data?*** Those techniques implemented in the pre-processing generate consistent high quality output images suitable for the deep learning and should be used. ***How well does the U-Net model perform at defect detection in ground-based overhead conductor images?*** Defects for each class are predicted in the images, just as the reported metrics suggest. Although the results showed that not all classes were predicted equally. ***How is the U-Net performance affected by factors like image resolution and the number of annotated images?*** Experiments showed that both image resolution and number of annotated images contribute to the performance of U-Net. The performance deteriorated at an increasing rate as the number of training images was reduced. Scaling down image resolution also reduced performance, although with mixed results for some classes.

In future research, attention will be given to quantity and precision of labelled images, as well as exploring how additional metadata can be used. Ideally, the overall algorithm will be integrated to a complete software package.

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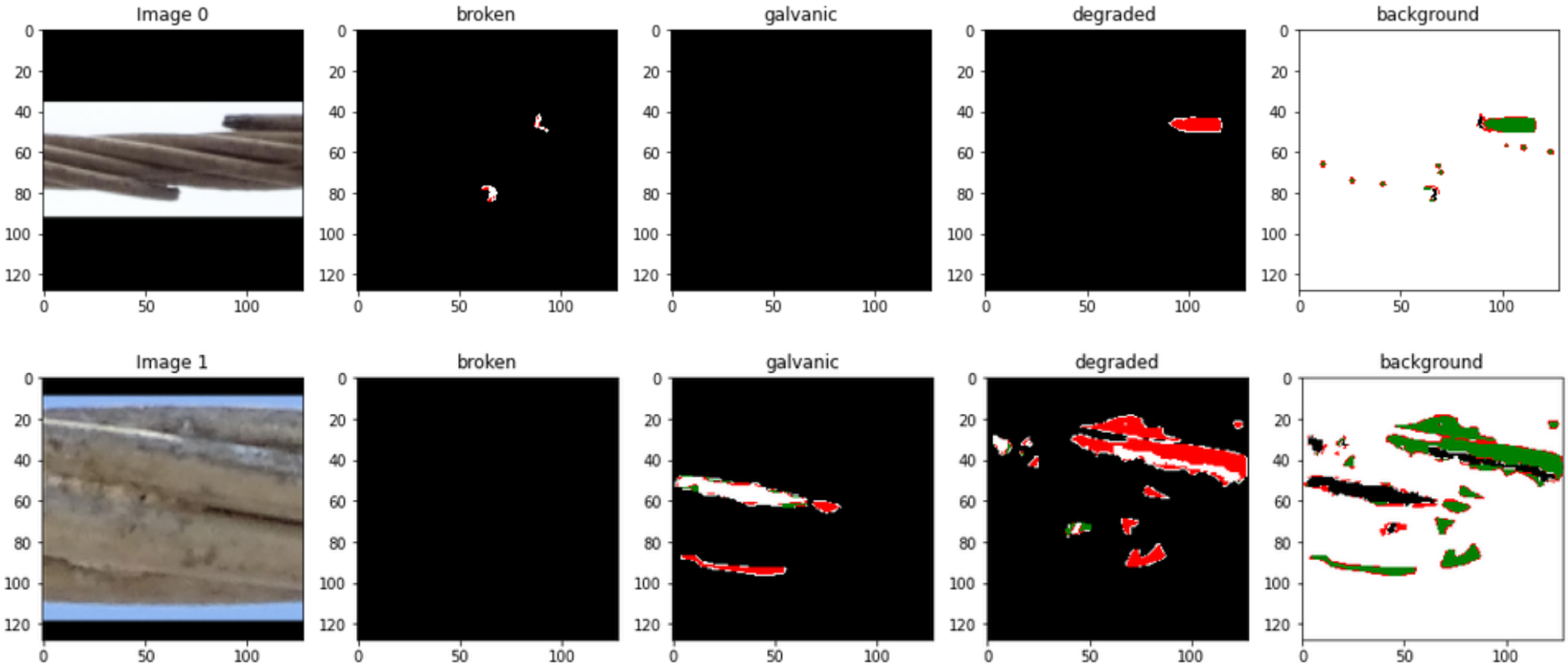
13 Appendix A

13.1 Inference on test data

The annotated masks in these images show large areas of false negatives, as they were early attempts at labelling and were rejected to be used only as test data. Some show overly large polygons and they overlap the extents of the defective areas.

Legend: `# black = background, white=TP, red=FN, green=FP`



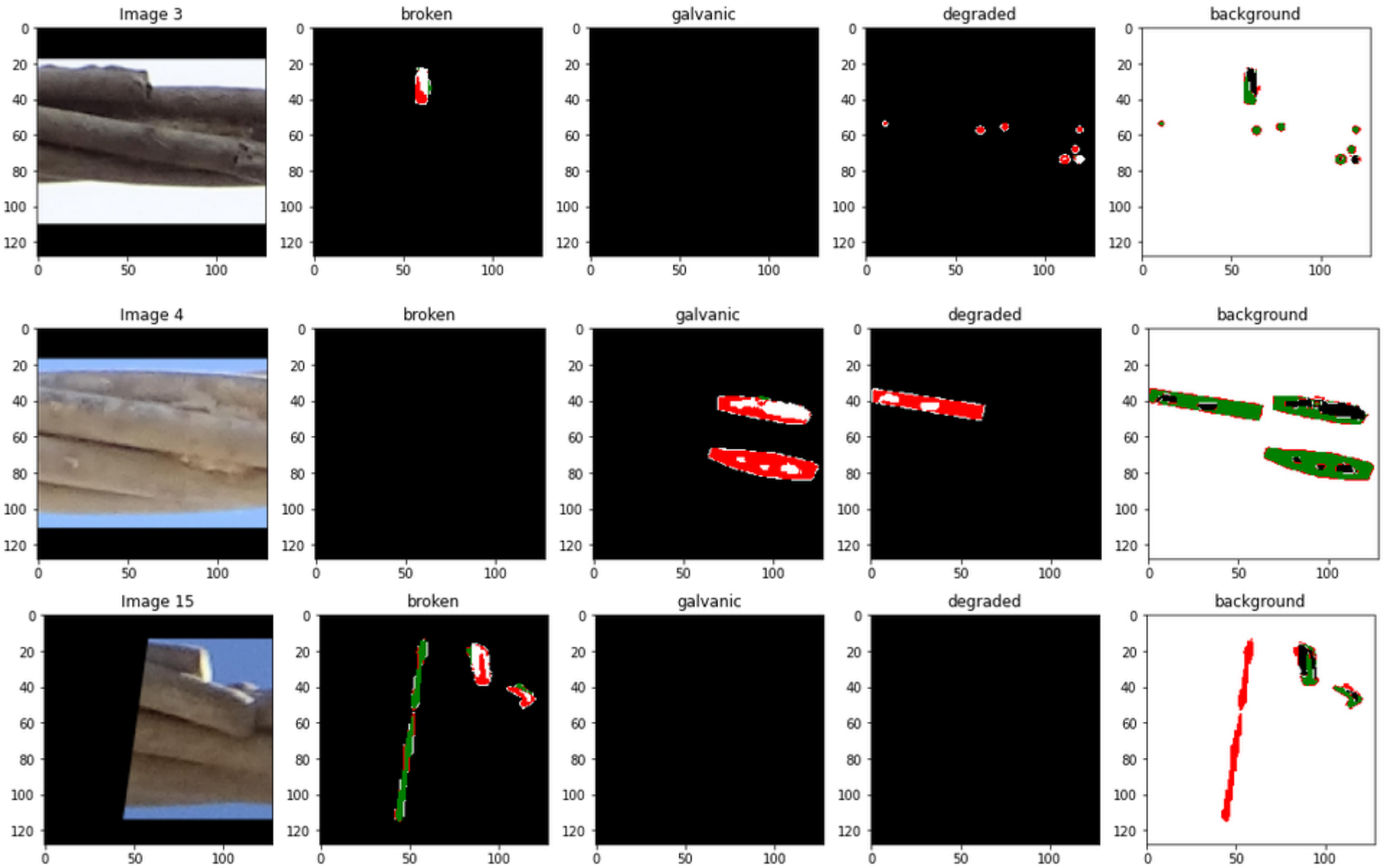


Image 3
broken
galvanic
degraded
background
Image 4
broken
galvanic
degraded
background
Image 15
broken
galvanic
degraded
background

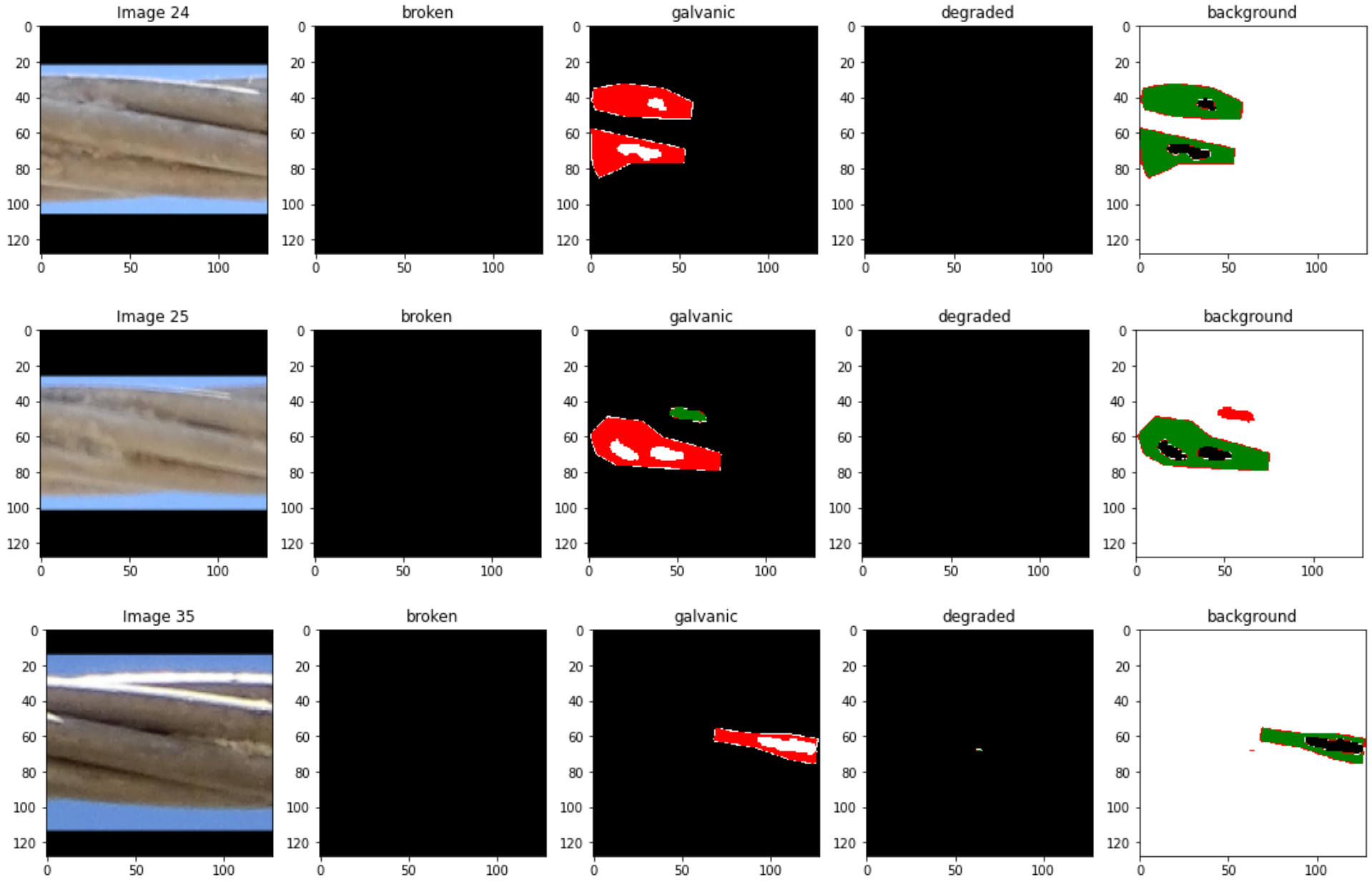
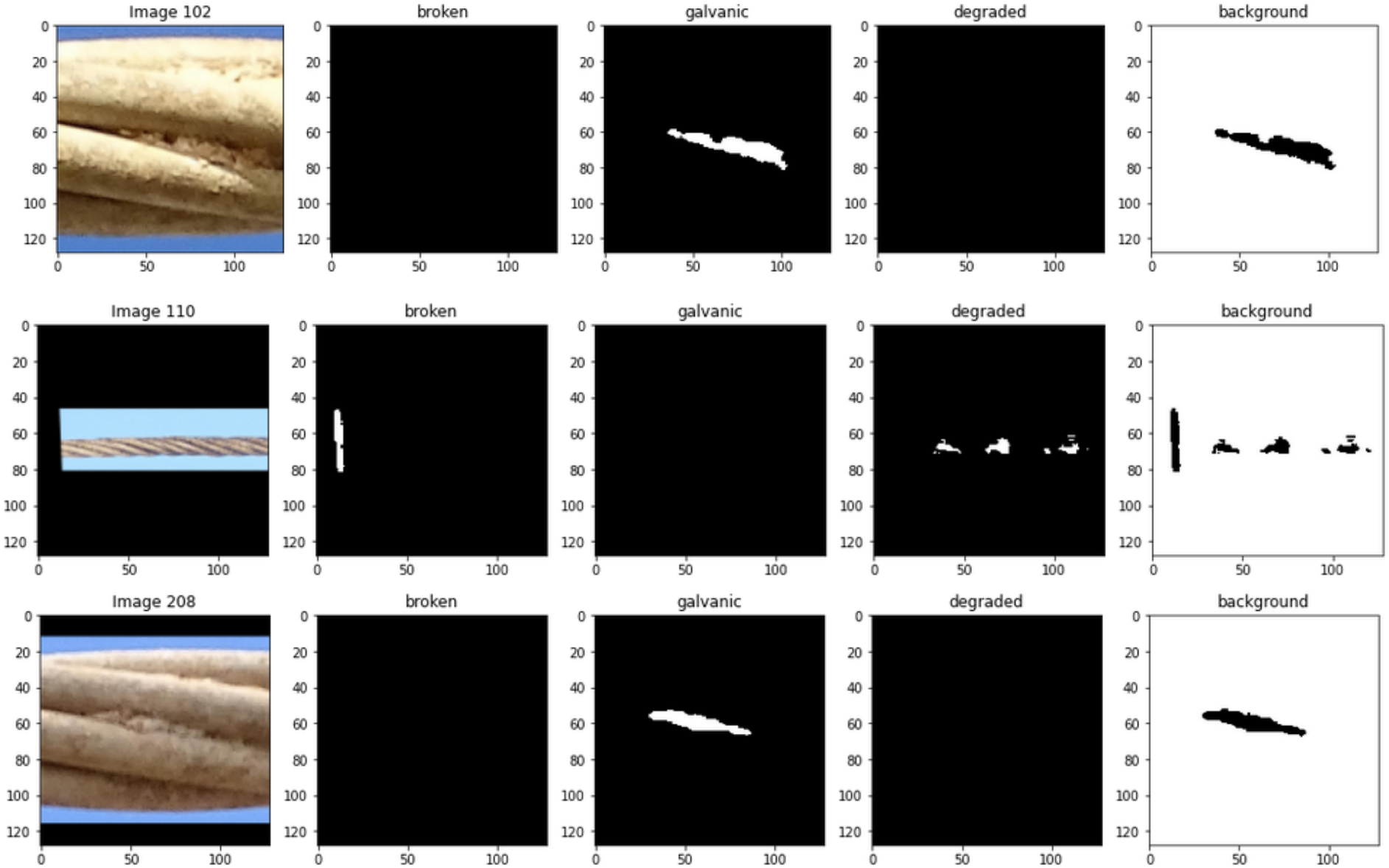


Image 24
broken
galvanic
degraded
background
Image 25
broken
galvanic
degraded
background
Image 35
broken
galvanic
degraded
background

13.2 Inference on unseen dev data

The images in this collection are selected from the initial data collection, they do not have masks and are treated only as unseen data.



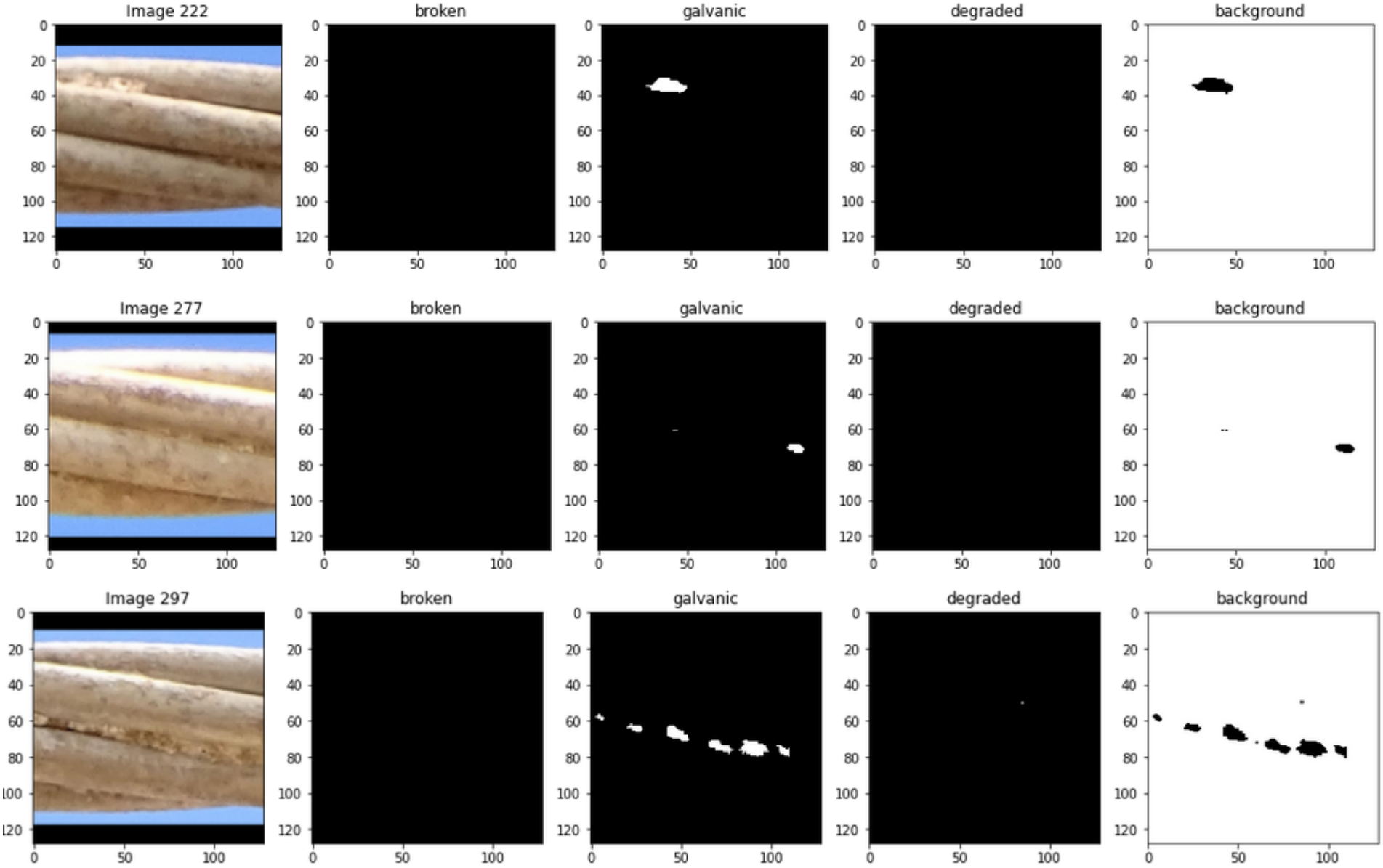


Image 222
broken
galvanic
degraded
background
Image 277
broken
galvanic
degraded
background
Image 297
broken
galvanic
degraded
background

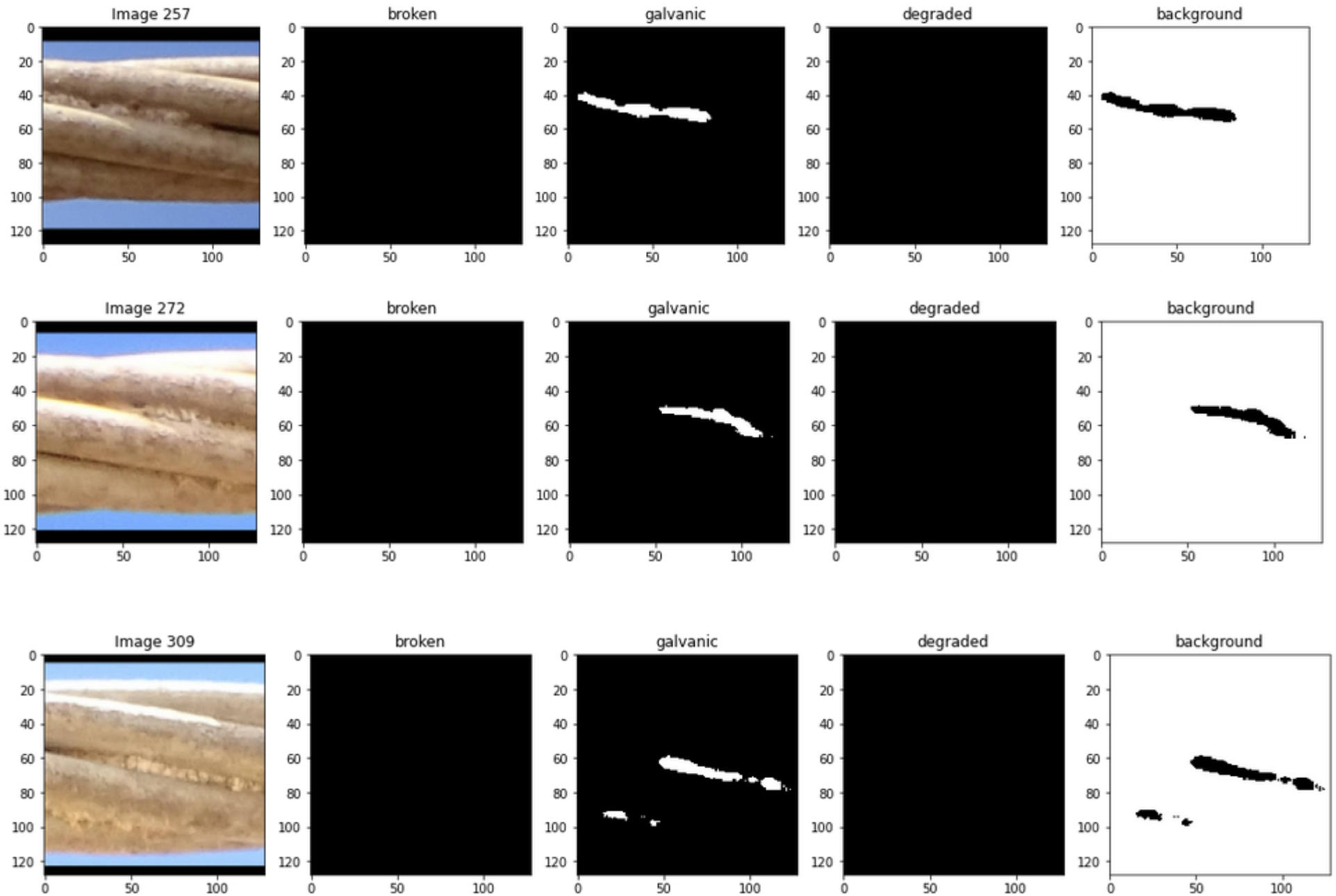
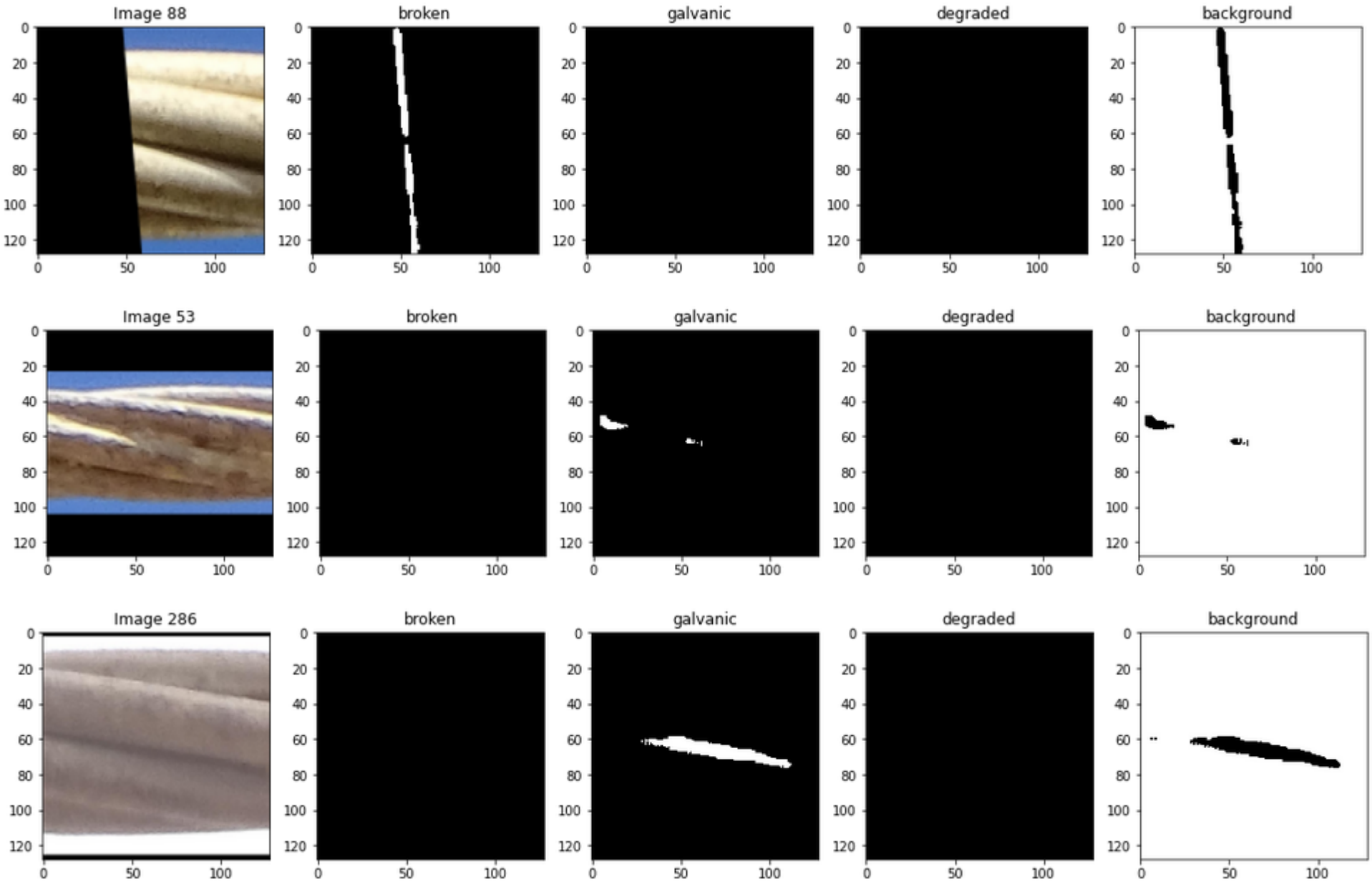


Image 257
broken
galvanic
degraded
background
Image 272
broken
galvanic
degraded
background
Image 309
broken
galvanic
degraded
background

13.3 Poor predictions examples